CSE 4615:: Wireless Networks

Enhancing Insight into
Wireless Transmission Fundamentals

Lecture 4

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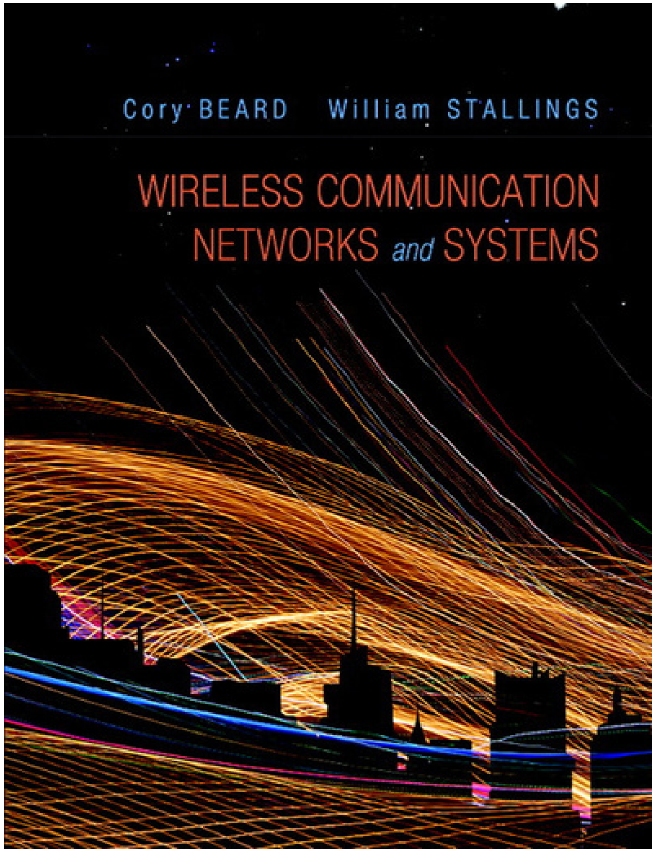
CHAPTER 2

TRANSMISSION FUNDAMENTALS



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CHAPTER 5 & 6 & 7
THE WIRELESS CHANNEL

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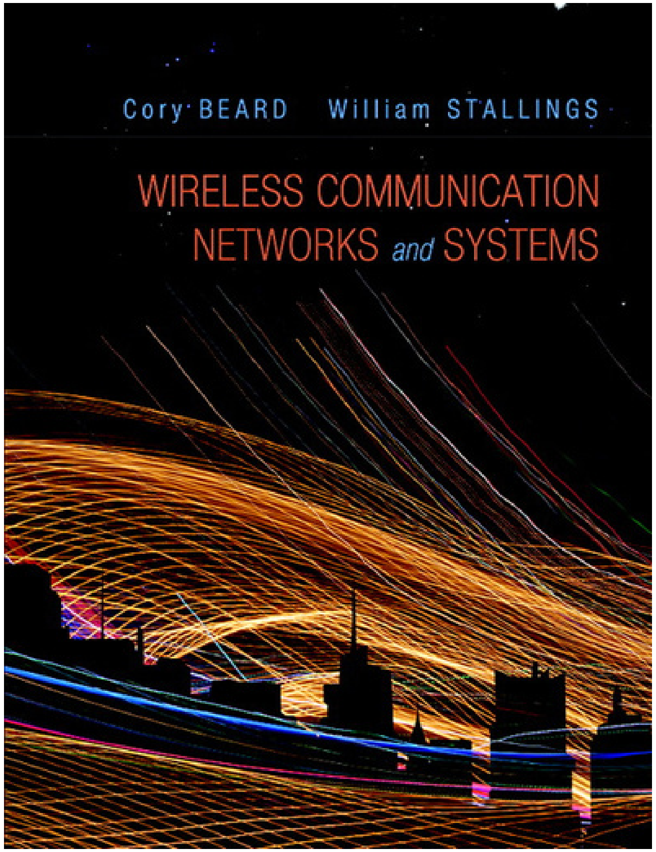

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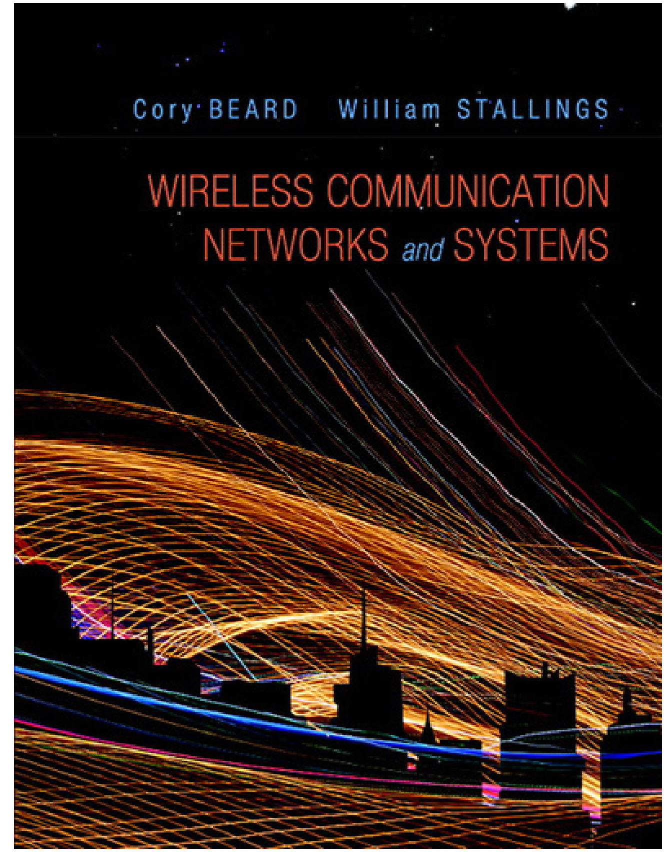
CHAPTER 8

ORTHOGONAL FREQUENCY DIVISION MULTIPLEXING



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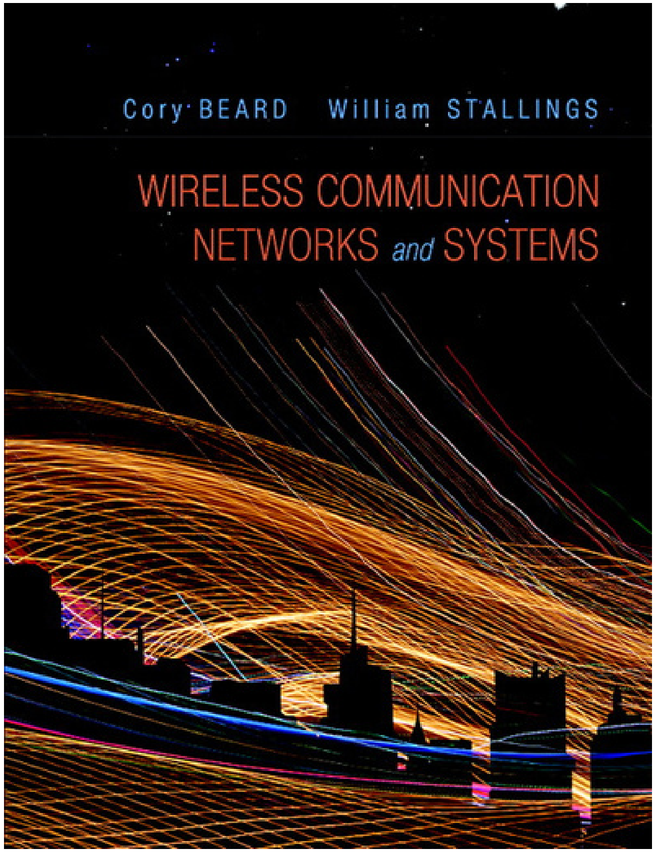
CHAPTER 9

SPREAD SPECTRUM



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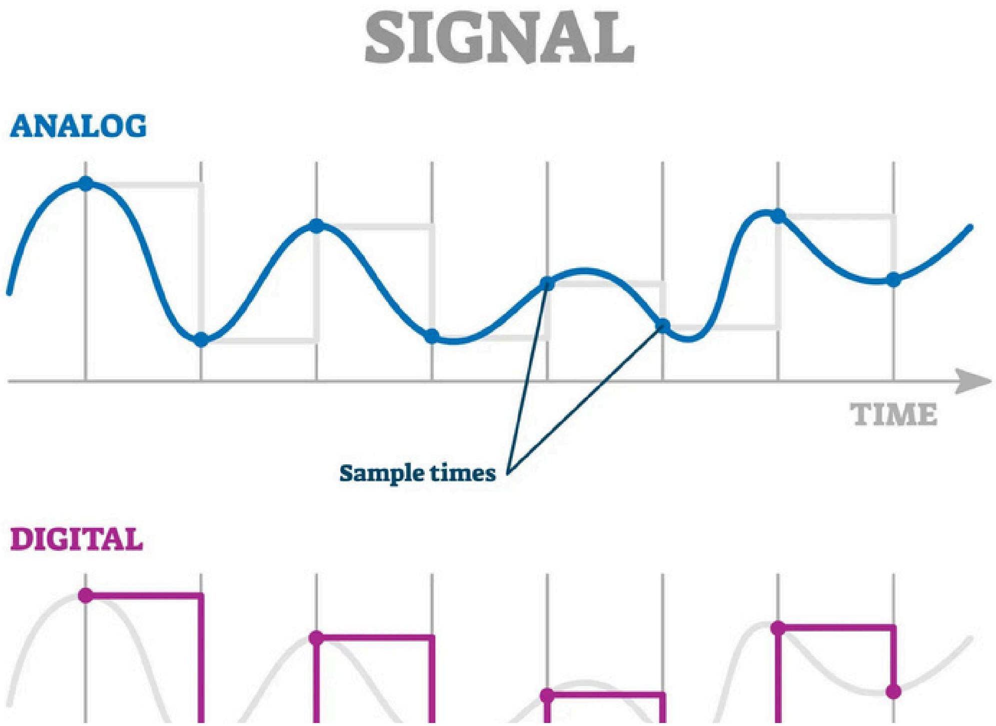
Electromagnetic Signal

Electromagnetic Signal

- Function of time
- Can also be expressed as a function of frequency
  - Signal consists of components of different frequencies

Electromagnetic Signal:: Time-Domain Concepts

- Analog signal - signal intensity varies in a smooth fashion over time
  - No breaks or discontinuities in the signal
- Digital signal - signal intensity maintains a constant level for some period of time and then changes to another constant level



Electromagnetic Signal::

Time-Domain Concepts

Electromagnetic Signal::
Time-Domain Concepts

- Aperiodic signal - analog or digital signal pattern that doesn't repeat over time
- Peak amplitude (*A*) - maximum value or strength of the signal over time; typically measured in volts
- Frequency (*f*)
  - Rate, in cycles per second, or Hertz (Hz) at which the signal repeats

Electromagnetic Signal::
Time-Domain Concepts

- Period (T) - amount of time it takes for one repetition of the signal
  - $T = 1/f$
- Phase (ϕ) - measure of the relative position in time within a single period of a signal
- Wavelength (λ) - distance occupied by a single cycle of the signal
  - Or, the distance between two points of corresponding phase of two consecutive cycles

2.2 Examples of Periodic Signals

Electromagnetic Signal::
Time-Domain Concepts

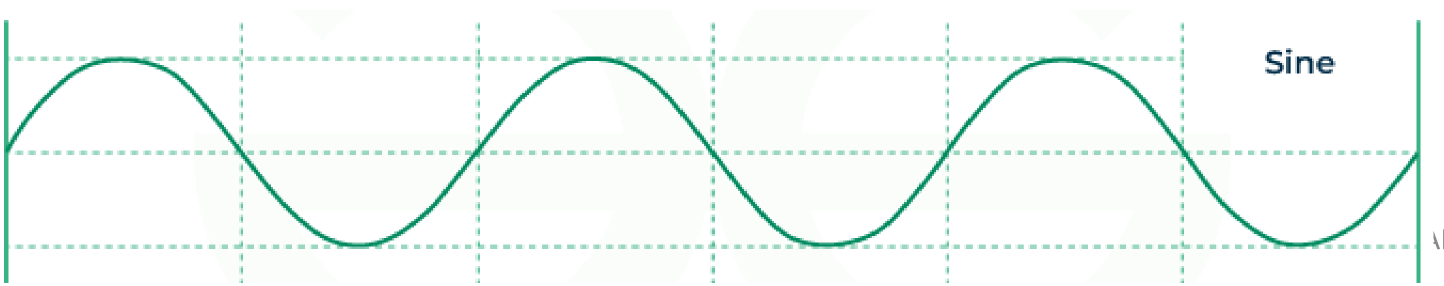
- Periodic signal - analog or digital signal pattern that repeats over time

$$s(t+T) = s(t) \quad -\infty < t < +\infty$$

  - where T is the period of the signal

- Types of Periodic Signals
  - Sinusoidal Wave
  - Square Wave
  - [Triangular Wave]
  - Sawtooth Wave
- Sinusoidal Wave
  - shape in the form of sine wave as shown below.
  - most common signal used in communication

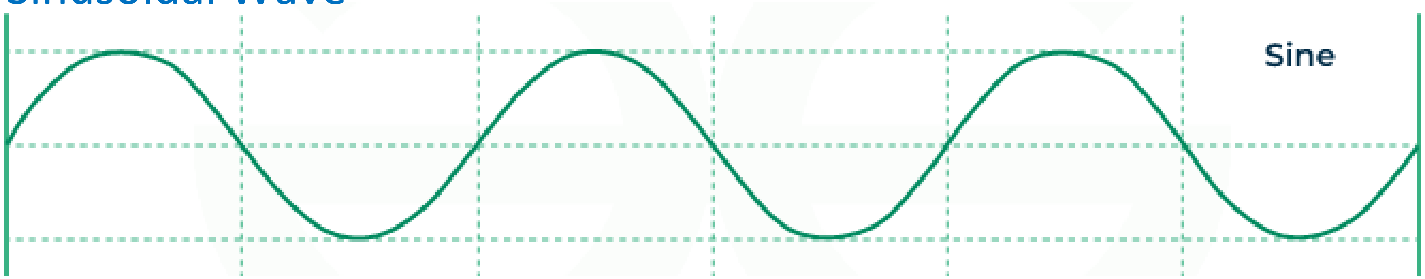
$$s(t)=A \cdot \sin(2\pi ft+\varphi)$$



Electromagnetic Signal:

Time-Domain Concepts

Sinusoidal Wave



$$s(t)=A \cdot \sin(2\pi ft+\varphi)$$

Where:

- **A** = amplitude (height of the wave)
- **f** = frequency (how fast it oscillates)
- **t** = time
- **ϕ** = phase (starting position of the wave)

The **phase** determines **where the sine wave starts** on the time axis.

φ = **Phase shift**, measured in **radians** or **degrees**

Phase Shift	What Happens
0° or 0 rad	Normal sine wave starts at 0 going up
90° or π/2 rad	Starts at the wave's peak
180° or π rad	Starts at 0 going **down**
270° or 3π/2 rad	Starts at lowest point

2.3 $s(t) = A \sin(2\pi f t + \varphi)$

Sine Wave Parameters

- General sine wave
  - $s(t) = A \sin(2\pi f t + \phi)$
- Figure 2.3 shows the effect of varying each of the three parameters
  - (a) $A = 1$, $f = 1$ Hz, $\phi = 0$; thus $T = 1$ s
  - (b) Reduced peak amplitude; A=0.5
  - (c) Increased frequency; $f = 2$, thus $T = ½$
  - (d) Phase shift; $\phi = \pi/4$ radians (45 degrees)
- Note: 2π radians = 360° = 1 period

Time vs. Distance

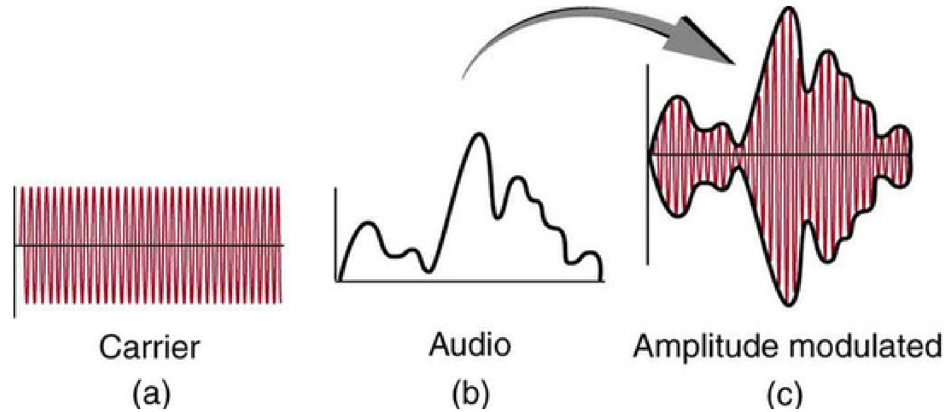
- When the horizontal axis is *time*, as in Figure 2.3, graphs display the value of a signal at a given point in *space* as a function of *time*
- With the horizontal axis in *space*, graphs display the value of a signal at a given point in *time* as a function of *distance*
  - At a particular instant of time, the intensity of the signal varies as a function of distance from the source

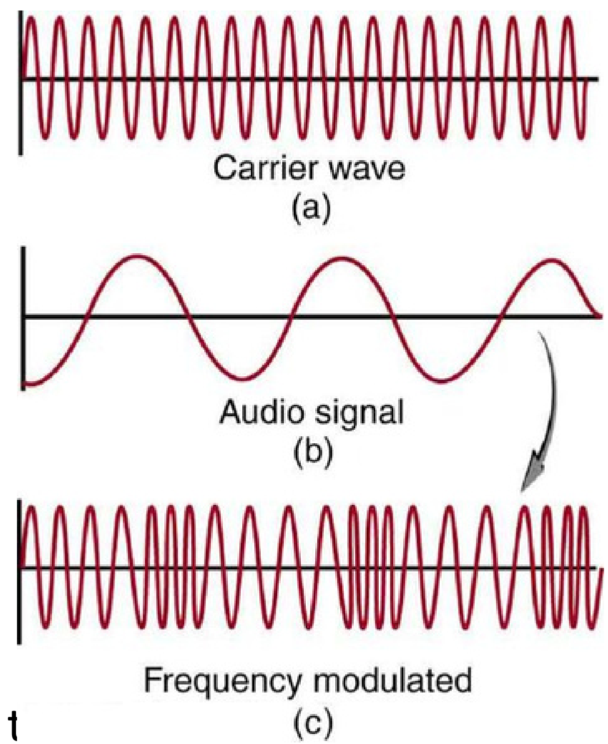
Electromagnetic Signal..

Time-Domain Concepts

In Communication:

- A **carrier signal** is usually a sine wave.
- We **modulate** that sine wave by changing:
  - **Amplitude** → Amplitude Modulation (AM)





  - **Frequency** → Frequency Modulation (FM)
  - **Phase** → Phase Modulation (PM)
- OFDM uses **multiple sine waves (subcarriers)**, each orthogonal t

Electromagnetic Signal:: Frequency-Domain Concepts

- Any electromagnetic signal can be shown to consist of a collection of periodic analog signals (sine waves) at different amplitudes, frequencies, and phases
- The period of the total signal is equal to the period of the fundamental frequency

Electromagnetic Signal:: Frequency-Domain Concepts

- Fundamental frequency - when all frequency components of a signal are integer multiples of one frequency, it's referred to as the fundamental frequency

Example

Suppose a signal is made of these frequency components:

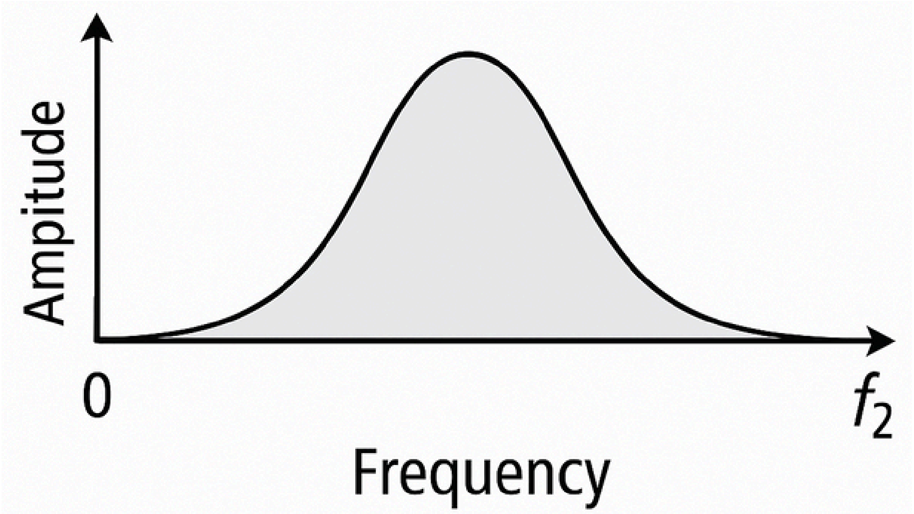
  - 50 Hz
  - 100 Hz
  - 150 Hz
  - 200 Hz
  - **So, 50 Hz is the fundamental frequency**

    In systems like :

  - Knowing the **fundamental frequency** helps understand the **structure and periodicity** of the signal.
  - It simplifies design (e.g., subcarrier spacing in OFDM is often based on a fundamental frequency).

Electromagnetic Signal::
Frequency-Domain Concepts

- Spectrum - range of frequencies that a signal contains



Electromagnetic Signal:: Frequency-Domain Concepts

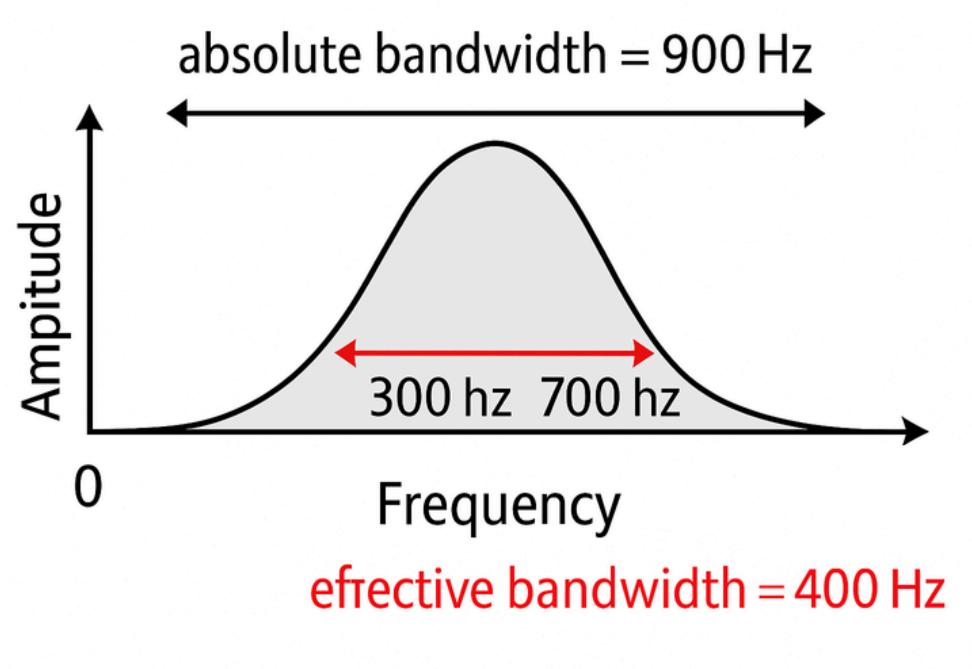
- Absolute bandwidth –
  - width of the spectrum of a signal
- Effective bandwidth (or just bandwidth) –
  - narrow band of frequencies that most of the signal's energy is contained in
  - Focuses on where the signal is **strongest**.
- Example:

  Suppose a signal spans:
  - From 100 Hz to 1000 Hz (absolute bandwidth = 900 Hz)
  - But 90% of the energy is between 300 Hz to 700 Hz → effective bandwidth = 400 Hz

Electromagnetic Signal:: Frequency-Domain Concepts

- Example:

  Suppose a sigr
  - From 100
  - But 90% (
    400 Hz



ctive bandwidth =

2.4 Addition of frequency Components(T = 1/*f*)

2.5 Frequency Components of Square Wave

Relationship between Data Rate and Bandwidth

- The greater the bandwidth, the higher the information-carrying capacity
- Conclusions
  - Any digital waveform will have many high-frequency components.
  - But in wireless, you can't transmit those infinite frequencies due to **antenna**, **RF front end**, and the **regulatory spectrum allocation**.
  - More bandwidth → higher power consumption, more **spectrum license fees**, more **complex hardware** (e.g., RFICs, ADCs)

Relationship between Data Rate and Bandwidth

- Conclusions
  - Wideband receivers are **costlier and more power-hungry** (important for phones, IoT devices).
  - in **multipath environments**, not all frequency components reach the receiver equally → **signal distortion** and errors.

Summary:

In wireless systems, **bandwidth is everything**:

✔ More bandwidth → higher data rates

✖ But it's **costly**, **regulated**, and hard to handle

✖ Limiting it can **degrade performance** due to distortion and loss of key signal components

Technology	Bandwidth	Max Data Rate
4G LTE	~20 MHz	~100 Mbps
5G mmWave	400 MHz+	1–10 Gbps

Data Communication Terms

Data Communication Terms

- Data - entities that convey meaning, or information
- Signals - electric or electromagnetic representations of data
- Transmission - communication of data by the propagation and processing of signals

Examples of Analog and Digital Data

- Analog
  - Audio
- Digital
  - Text
  - Integers

Analog Signals

- A continuously varying electromagnetic wave that may be propagated over a variety of media, depending on frequency
- Examples of media:
  - Copper wire media (twisted pair and coaxial cable)
  - Fiber optic cable
  - Atmosphere or space propagation
- Analog signals can propagate analog and digital data

Digital Signals

- A sequence of voltage pulses that may be transmitted over a copper wire medium
- Generally cheaper than analog signaling
- Less susceptible to noise interference
- Digital signals can propagate analog and digital data

Digital Signals

Where Digital Signals Are Used in Communication Networks

Technology	Medium	Digital Signal Used Directly?	Notes
Ethernet (LAN)	Copper wire	✅ Yes	Voltage pulses
Fiber optics	Glass fiber	✅ Yes	As light pulses
Wi-Fi / Cellular	Air (RF)	❌ (modulated)	Uses analog RF wave
USB / PCIe / SATA	Copper traces	✅ Yes	Direct bit-level transfer

Reasons for Choosing Data and Signal Combinations

- Digital data, digital signal
  - Equipment for encoding is less expensive than digital-to-analog equipment
  - Common in wired LANs (Ethernet) and USB.
- Analog data, digital signal
  - Conversion permits use of modern digital transmission and switching equipment
  - Used in digital telephony
- Digital data, analog signal
  - Some media (like **radio waves** or **satellites**) only support **analog waveform propagation**.
  - Binary data is **modulated** into an **analog carrier wave** (e.g., using ASK, FSK, PSK, QAM).
  - Used in: Wi-Fi, 4G/5G (QAM), Satellite internet
- Analog data, analog signal
  - Analog information (like voice) is directly modulated into an **analog signal** (e.g., AM or FM).
  - Used in traditional radio broadcasting, analog telephony

2.8 Analog and Digital Signaling of Analog and Digital Data

Analog vs Digital Transmission

Feature	Analog Transmission	Digital Transmission
Signal Type	Continuous waveform (e.g., sine wave)	Discrete binary signals (0s and 1s)
Transmission Content	No regard to data content (signal is passed as-is)	Data is encoded and decoded in digital format
Attenuation Handling	Requires **amplifiers** to boost signal	Uses **repeaters** to regenerate the original digital signal
Distortion Handling	Analog signals **tolerate distortion** to some extent	Digital signals are **more sensitive** to distortion
Amplification Effect	Cascaded amplifiers also **amplify noise and distortion**	Digital repeaters **clean up** the signal
Noise Immunity	Lower – analog signal quality degrades with noise	Higher – digital thresholds help recover the original data

Channel Capacity

Channel Capacity

- Impairments, such as noise, limit data rate that can be achieved
- For digital data, to what extent do impairments limit data rate?
- Channel Capacity –

  the maximum rate at which data can be transmitted over a given communication path, or channel, under given conditions

2.9 Effect of Noise on Digital Signal

Concepts Related to Channel Capacity

- Data rate - rate at which data can be communicated (bps)
- Bandwidth - the bandwidth of the transmitted signal as constrained by the transmitter and the nature of the transmission medium (Hertz)
- Noise - average level of noise over the communications path
- Error rate - rate at which errors occur
  - Error = transmit 1 and receive 0; transmit 0 and receive 1

Classifications of Transmission Media

- Transmission Medium
  - Physical path between transmitter and receiver
- Guided Media
  - Waves are guided along a solid medium
  - E.g., copper twisted pair, copper coaxial cable, optical fiber
- Unguided Media
  - Provides means of transmission but does not guide electromagnetic signals
  - Usually referred to as wireless transmission
  - E.g., atmosphere, outer space

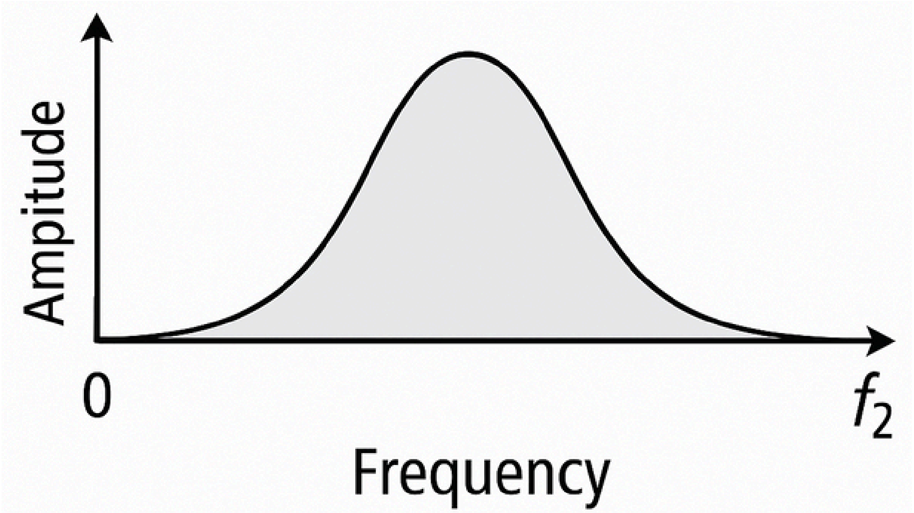
Unguided Media

- Transmission and reception are achieved by means of an antenna
- Configurations for wireless transmission
  - Directional
  - Omnidirectional

Frequency Spectrum

Frequency Spectrum

- Spectrum - range of frequencies that a signal contains



Spectrum considerations

- Controlled by regulatory bodies
  - Carrier frequency
  - Signal Power
  - Multiple Access Scheme
    - Divide into time slots –Time Division Multiple Access (TDMA)
    - Divide into frequency bands – Frequency Division Multiple Access (FDMA)
    - Different signal encodings – Code Division Multiple Access (CDMA)

Spectrum considerations

- Federal Communications Commission (FCC) in the United States regulates spectrum
  - Military
  - Broadcasting
  - Public Safety
  - Mobile
  - Amateur
  - Government exclusive, non-government exclusive, or both
  - Many other categories

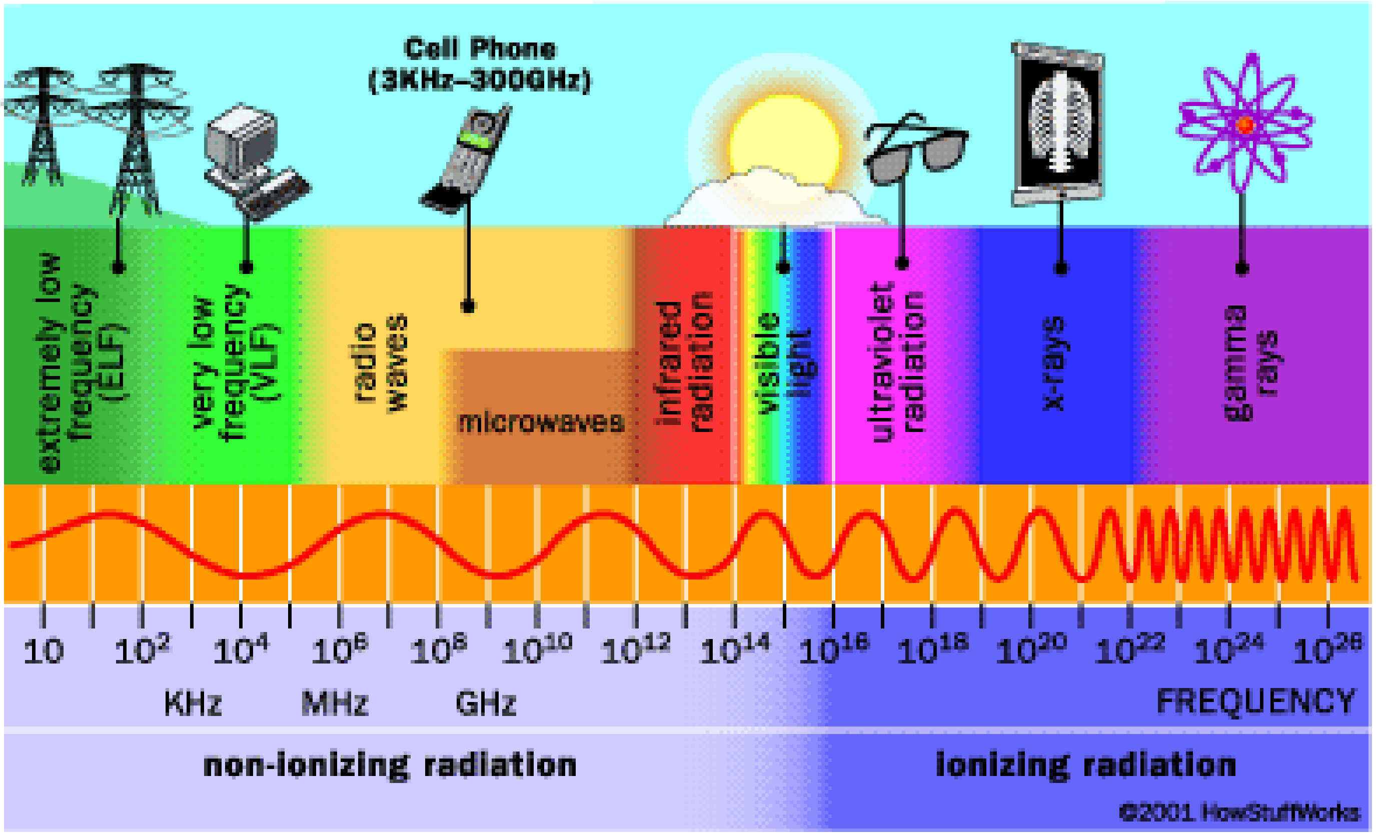
Spectrum considerations

- Industrial, Scientific, and Medical (ISM) bands
  - Can be used without a license
  - As long as power and spread spectrum regulations are followed
- ISM bands are used for
  - WLANs
  - Wireless Personal Area networks
  - Internet of Things

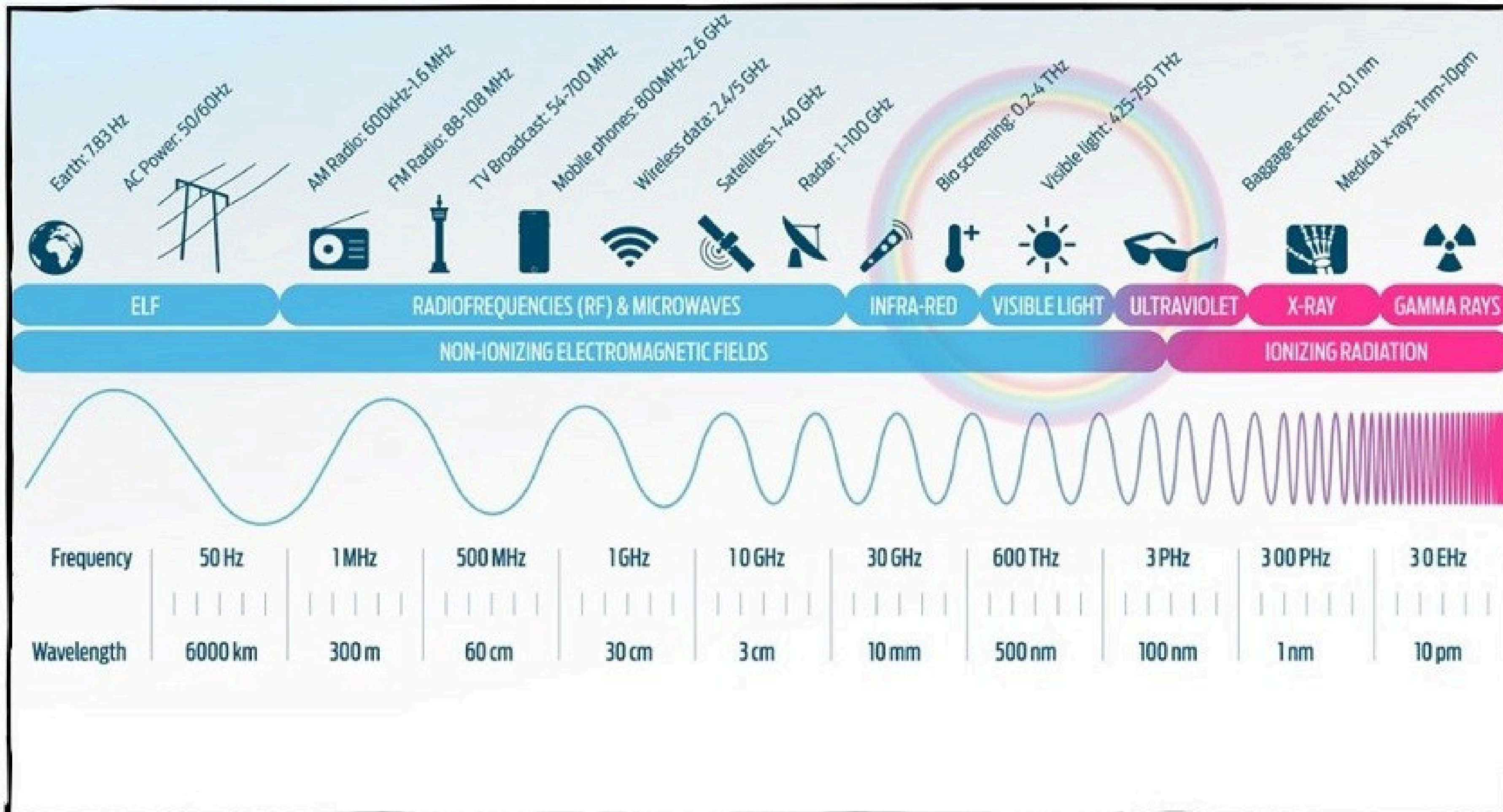
Frequency Spectrum

2.10 Electromagnetic spectrum of Telecommunications

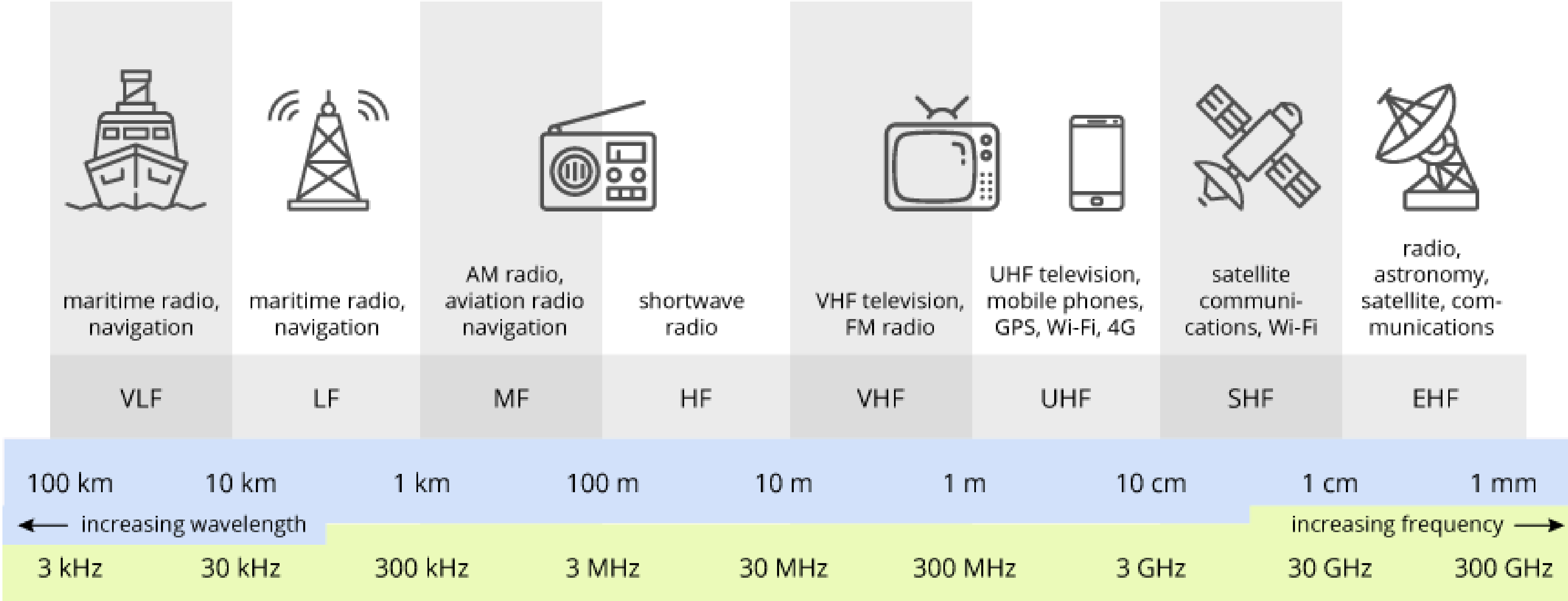
Frequency Spectrum



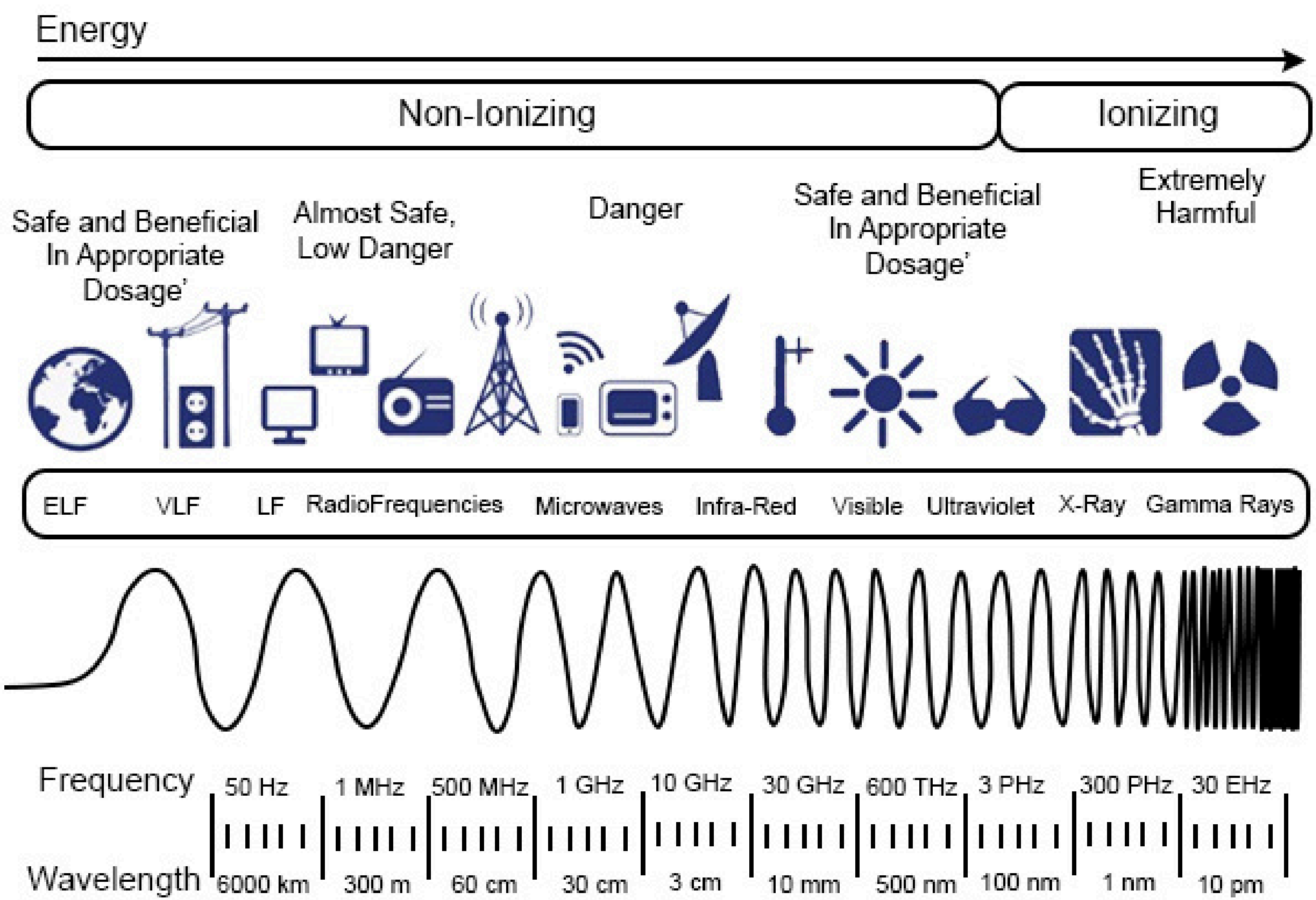
Frequency Spectrum



Frequency Spectrum



Frequency Spectrum



Wireless Transmission Frequencies

- 3×10^{11} to 2×10^{14}
  - Infrared
  - Local
  - Line of sight (or reflection)
  - Blocked by walls
  - e.g. TV remote control, IRD port
- 1GHz to 30GHz
  - Microwave
  - Highly directional
  - Point to point
  - Satellite
- 30MHz to 1GHz
  - Omnidirectional
  - Broadcast radio

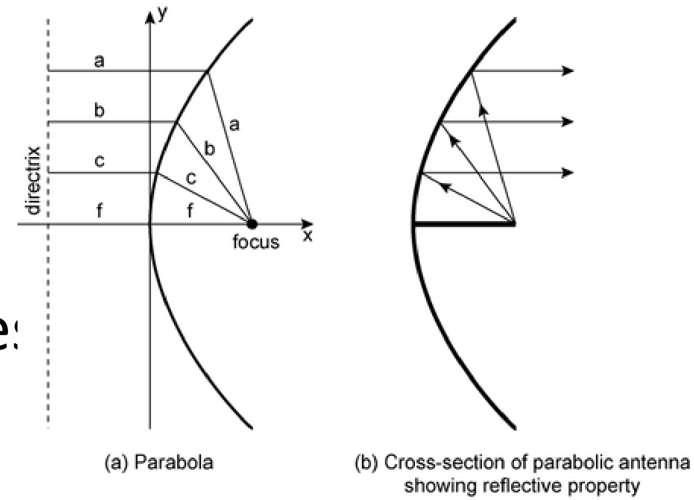
Types of Wireless Transmission on Wireless Channel::

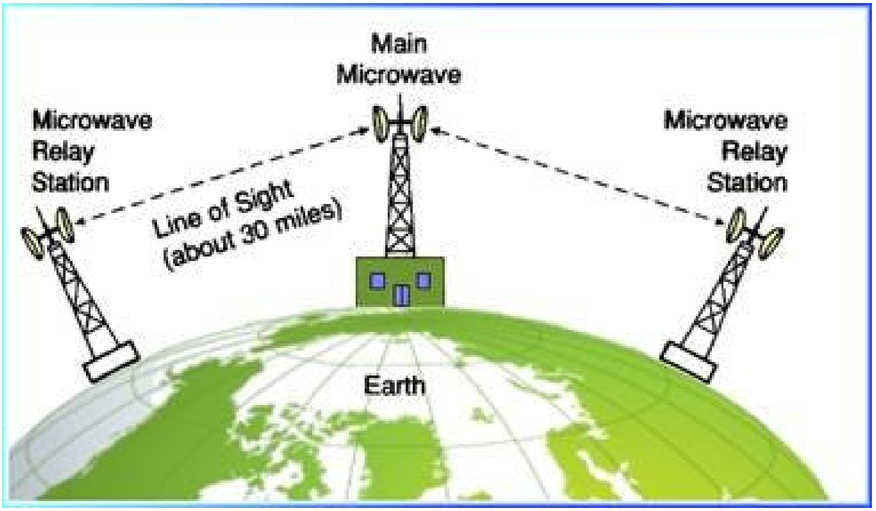
Radio and Microwave Transmission

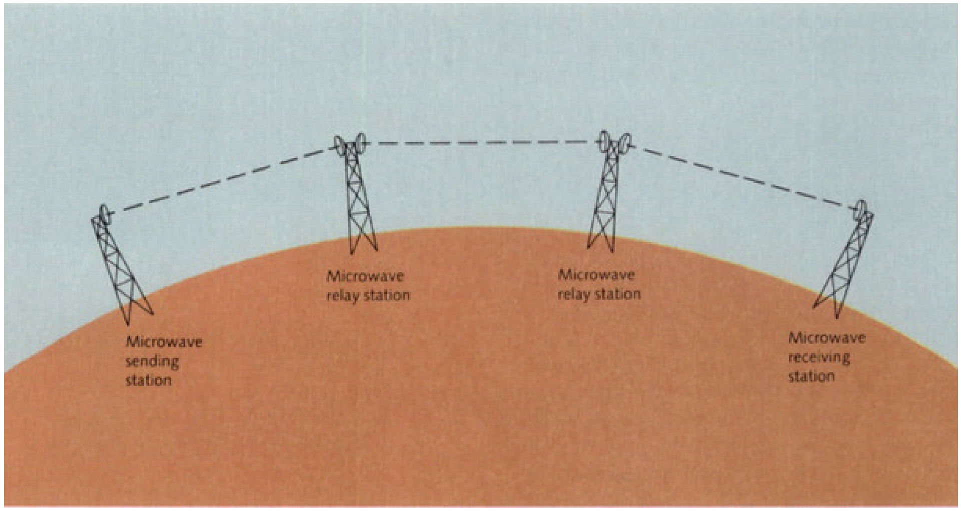
Type	Medium	Direction	Frequency Range	Description & Use
1. Terrestrial Microwave Transmission	Earth-to-Earth	Line-of-sight	1–30 GHz	• Highly directional • Point to point
2. Satellite Microwave Transmission	Earth-to-Satellite	Long-range	Uplink: 5.9–6.4 GHz Downlink: 3.7–4.2 GHz	• Long-range communication via satellites (TV, GPS)
3. Broadcast Radio Transmission	Tower-to-Many	Omnidirectional	AM: 535–1605 kHz FM: 88–108 MHz	• Mass public communication for entertainment & alerts

1. Terrestrial Microwave

- Parabolic dish
- Focused beam
- Line of sight
- Long haul telecommunications
- Higher frequencies give higher data rates







1. Terrestrial Microwave

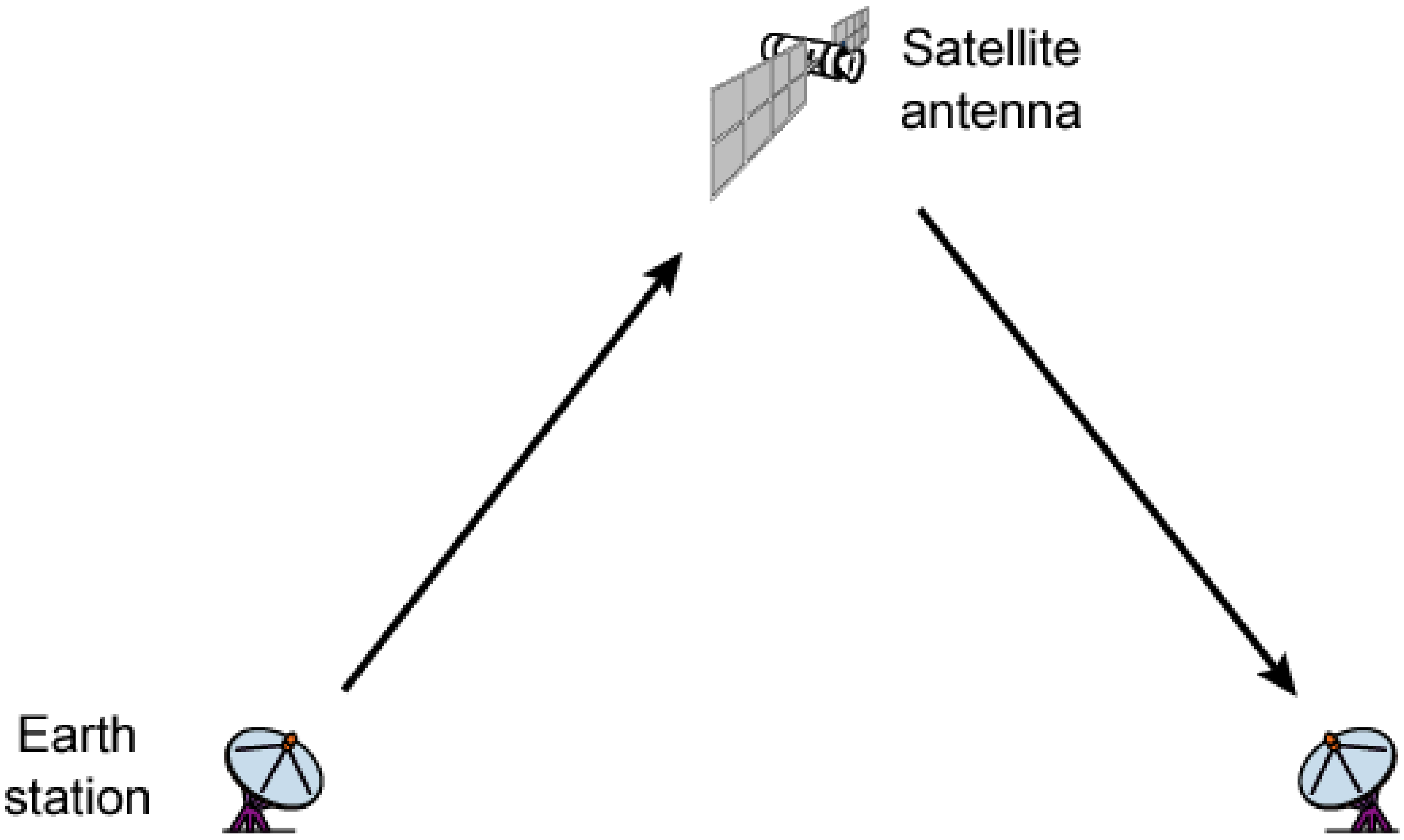
Definition:
Microwave communication between **two ground-based antennas**, typically mounted on towers or buildings.
Frequency Range:
Usually between **1 GHz to 30 GHz**.
Description of Common Microwave Antenna:
- Uses a **parabolic "dish"**, around **3 meters in diameter**.
- Antennas are **rigidly fixed** and **focus a narrow beam**.
- Requires **line-of-sight** transmission to the receiving antenna.
- Antennas are mounted at **substantial heights above ground level** to avoid obstructions.

How it works:
- Line-of-sight transmission ensures high-speed communication with **low attenuation** over long distances.

Applications:
- Long-haul telecommunications services (e.g., inter-city links, WiMAX [Worldwide Interoperability for Microwave Access]).
- Short **point-to-point links between buildings**, such as for corporate or campus networks.

2. Satellite Microwave

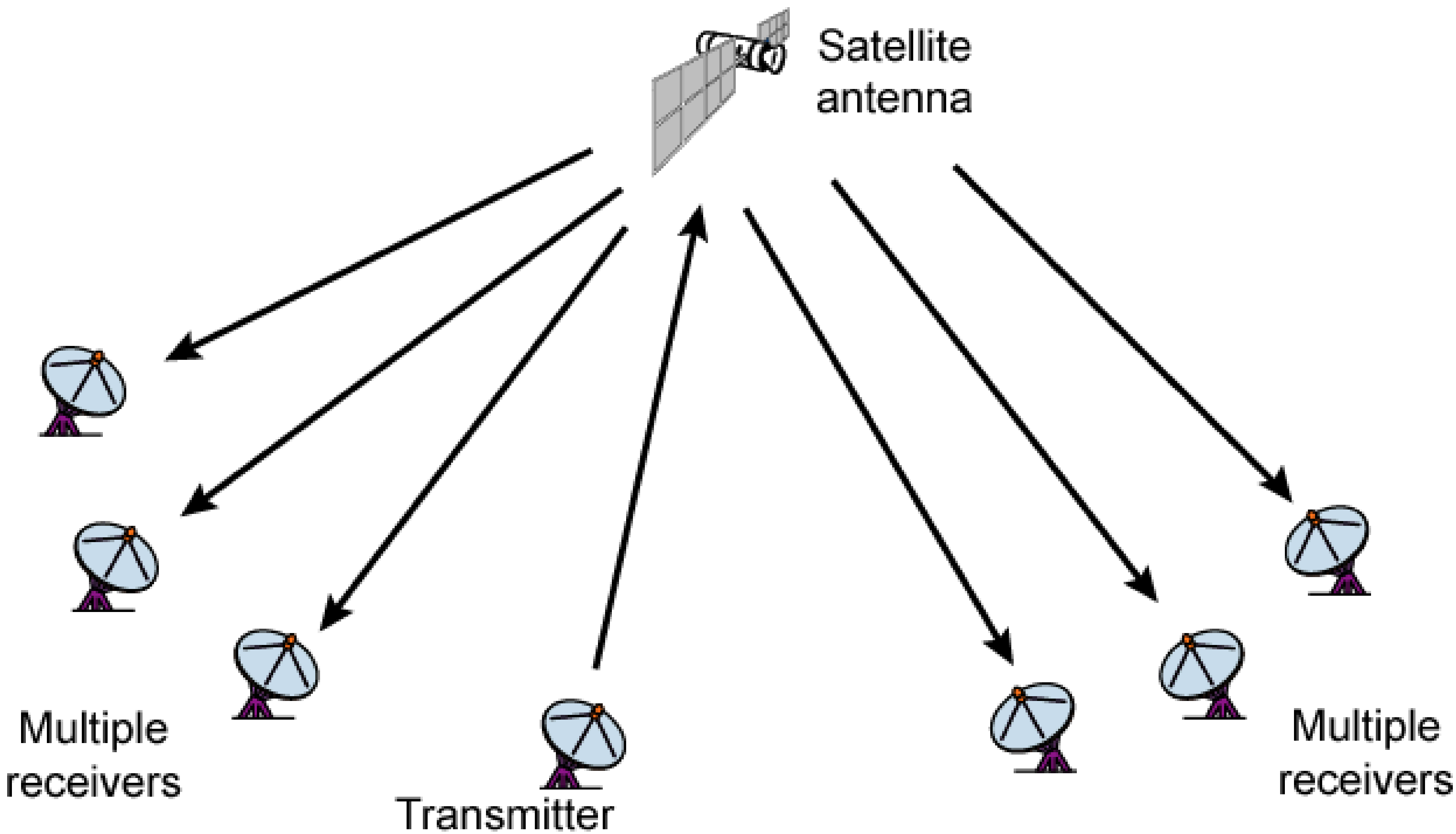
Satellite Point to Point Link



(a) Point-to-point link

2. Satellite Microwave

Satellite Broadcast Link



(b) Broadcast link

2. Satellite Microwave

- Satellite is relay station
- Satellite receives on one frequency, amplifies or repeats signal and transmits on another frequency
- Requires geo-stationary orbit
  - Height of 35,784km
- Television
- Long distance telephone
- Private business networks

Satellite Microwave

Definition:
Microwave communication **between Earth stations and satellites** orbiting the Earth.
Frequency Range:

- **Uplink** (Earth to satellite): ~5.9–6.4 GHz
- **Downlink** (Satellite to Earth): ~3.7–4.2 GHz
  (*Ranges vary by band – C, Ku, Ka, etc.*)
  How it works:
- Ground station transmits to a satellite (**uplink**).
- Satellite retransmits the signal to another ground station (**downlink**).
- Useful for **global coverage**, even in remote areas.
  Applications ?:
- **TV broadcasting**
- **GPS and navigation**
- **Internet in remote areas**
- **International voice and data communication**

3. Broadcast Radio

- Omnidirectional
- FM radio
- UHF and VHF television
- Line of sight
- Suffers from multipath interference
  - Reflections

3. Broadcast Radio

Definition:
Traditional radio communication from **a single broadcasting tower** to **many receivers**.
Frequency Range:
- **AM radio**: 535–1605 kHz
- **FM radio**: 88–108 MHz

How it works:
- A **central antenna** broadcasts signals in an **omnidirectional pattern**.
- Can cover **large geographic areas** depending on frequency and power.

Applications:
- **AM/FM radio services**
- **Emergency broadcasts**
- **Public announcements**
- **Mass public communication for entertainment & alerts**

3. Broadcast Radio

- **AM/FM radio services**

Feature	**AM (Amplitude Modulation)**	**FM (Frequency Modulation)**
Full Name	Amplitude Modulation	Frequency Modulation
Carrier Frequency Band	535 – 1605 kHz (kilohertz)	88 – 108 MHz (megahertz)
Modulation	Information is encoded by **varying amplitude** of the carrier wave	Information is encoded by **varying frequency** of the carrier wave
Sound Quality	Lower (more noise-sensitive)	Higher (better resistance to noise)

Why the Difference?

- **AM** is **cheaper** and **travels farther**, but more affected by **noise** like lightning or electrical equipment.
- **FM** uses more bandwidth and is more **complex**, but delivers **clearer sound** and is **less prone to interference**.

Intuitive Explanation:

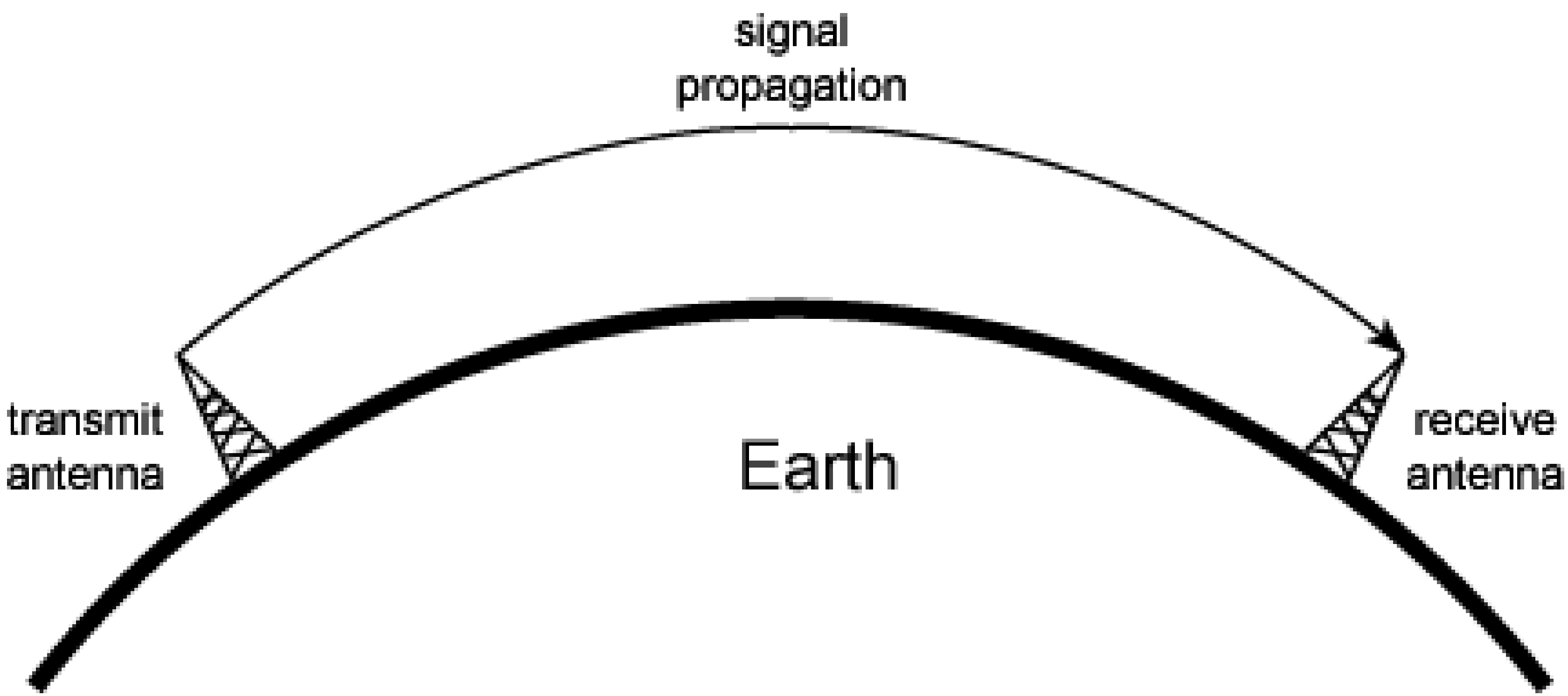
- AM is like **shouting louder or softer** (amplitude changes).
- FM is like **changing the pitch/tone** (frequency changes) to carry the same message.

Wireless Propagation Modes

- Signal travels along three routes
  - Ground wave
    - Follows contour of earth
    - Up to 2MHz
    - AM radio
  - Sky wave
    - Amateur radio, BBC world service, Voice of America
    - Signal reflected from ionosphere layer of upper atmosphere
    - (Actually refracted)
  - Line of sight
    - Above 30Mhz
    - May be further than optical line of sight due to refraction
    - More later…

Wireless Propagation Modes

Ground Wave Propagation

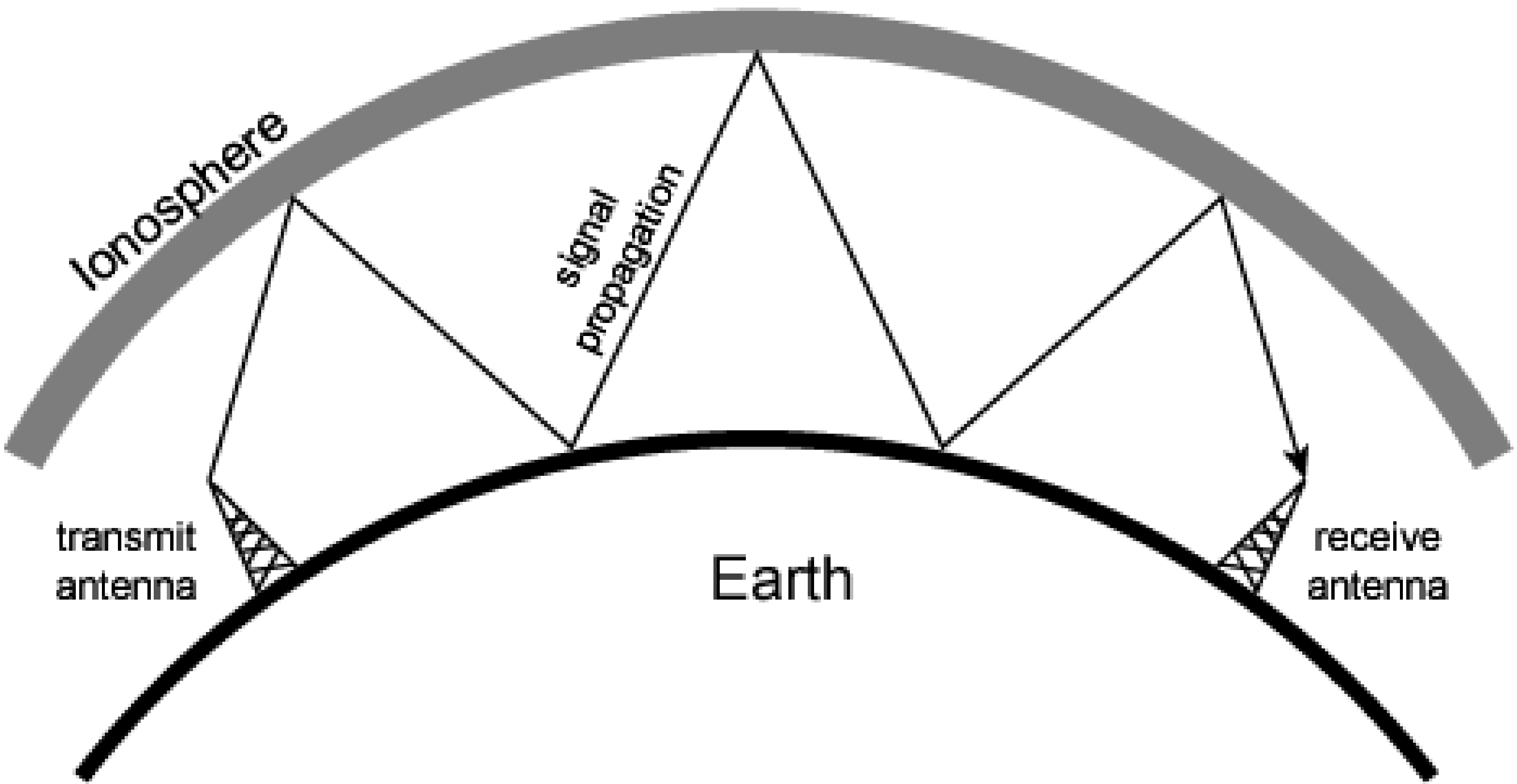


(a) Ground-wave propagation (below 2 MHz)

Ground Wave Propagation

- Follows contour of the earth
- Can propagate considerable distances
- Frequencies up to 2 MHz
- Example
  - AM radio

Sky Wave Propagation

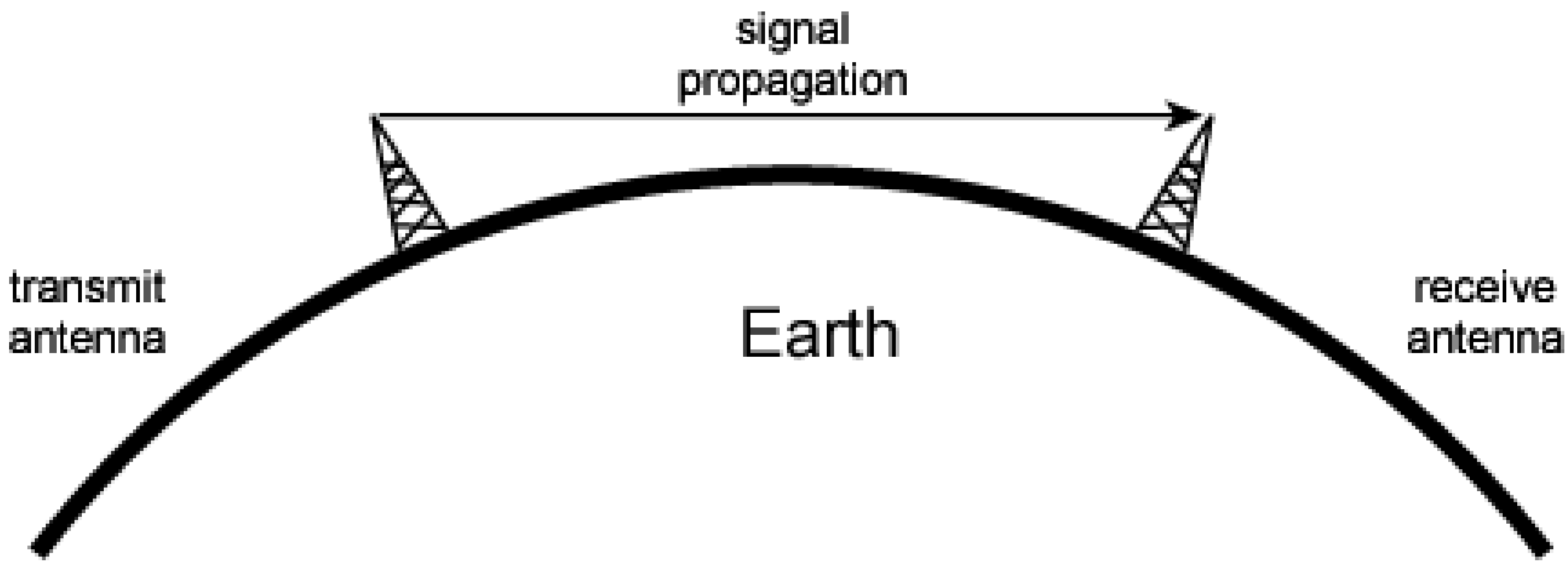


(b) Sky-wave propagation (2 to 30 MHz)

Sky Wave Propagation

- Signal reflected from ionized layer of atmosphere back down to earth
- Signal can travel a number of hops, back and forth between ionosphere and earth's surface
- Reflection effect caused by refraction
- Examples
  - Amateur radio
  - CB radio

Line of Sight Propagation



(c) Line-of-sight (LOS) propagation (above 30 MHz)

Line of Sight Transmission

- Free space loss
  - Signal disperses with distance
  - Greater for lower frequencies (longer wavelengths)
- Atmospheric Absorption
  - Water vapour and oxygen absorb radio signals
  - Water greatest at 22GHz, less below 15GHz
  - Oxygen greater at 60GHz, less below 30GHz
  - Rain and fog scatter radio waves
- Multipath
  - Better to get line of sight if possible
  - Signal can be reflected causing multiple copies to be received
  - May be no direct signal at all
  - May reinforce or cancel direct signal
- Refraction
  - May result in partial or total loss of signal at receiver

Line-of-Sight Propagation

- Transmitting and receiving antennas must be within line of sight
  - Satellite communication – signal above 30 MHz not reflected by ionosphere
  - Ground communication – antennas within effective line of site due to refraction
- Refraction – bending of microwaves by the atmosphere
  - Velocity of electromagnetic wave is a function of the density of the medium
  - When wave changes medium, speed changes
  - Wave bends at the boundary between mediums

The Effects of Multipath Propagation

- Reflection, diffraction, and scattering
- Multiple copies of a signal may arrive at different phases
  - If phases add destructively, the signal level relative to noise declines, making detection more difficult
- Intersymbol interference (ISI)
  - One or more delayed copies of a pulse may arrive at the same time as the primary pulse for a subsequent bit
- Rapid signal fluctuations
  - Over a few centimeters

Propagation Mechanisms

1. Free-space propagation
2. Transmission
   - Through a medium
   - Refraction occurs at boundaries
3. Reflections
   - Waves impinge upon surfaces that are large compared to the signal wavelength
4. Diffraction
   - Secondary waves behind objects with sharp edges
5. Scattering
   - Interactions between small objects or rough surfaces

Propagation Mechanisms

Sketch of Three Important Propagation Mechanisms

R= Reflections
D = Diffraction
S = Scattering

Antennas

- An antenna is an electrical conductor or system of conductors
  - Transmission - radiates electromagnetic energy into space
  - Reception - collects electromagnetic energy from space
- In two-way communication, the same antenna can be used for transmission and reception but not at the same time.

Attenuation

- Strength of signal falls off with distance over transmission medium
- Attenuation factors for unguided media:
  - Received signal must have sufficient strength so that circuitry in the receiver can interpret the signal
  - Signal must maintain a level sufficiently higher than noise to be received without error
  - Attenuation is greater at higher frequencies, causing distortion

Loss Models Derived from Empirical Measurements

- Need to design systems based on empirical data applied to a particular environment
  - To determine power levels, tower heights, height of mobile antennas
- Okumura developed a model, later refined by Hata
  - Detailed measurement and analysis of the Tokyo area
  - Among the best accuracy in a wide variety of situations
- Predicts path loss for typical environments
  - Urban
  - Small, medium sized city
  - Large city
  - Suburban
  - Rural

Categories of Noise

- Thermal Noise
- Intermodulation noise
- Crosstalk
- Impulse Noise

Thermal Noise

- Thermal noise due to agitation of electrons
- Present in all electronic devices and transmission media
- Cannot be eliminated
- Function of temperature
- Particularly significant for satellite communication

Noise Terminology

- Intermodulation noise – occurs if signals with different frequencies share the same medium
  - Interference caused by a signal produced at a frequency that is the sum or difference of original frequencies
- Crosstalk – unwanted coupling between signal paths
- Impulse noise – irregular pulses or noise spikes
  - Short duration and of relatively high amplitude
  - Caused by external electromagnetic disturbances, or faults and flaws in the communications system

Other Impairments

- Atmospheric absorption – water vapor and oxygen contribute to attenuation
- Multipath – obstacles reflect signals so that multiple copies with varying delays are received
- Refraction – bending of radio waves as they propagate through the atmosphere

The Trouble with Wireless

The trouble with wireless

- Wireless is convenient and less expensive, but not perfect
- Limitations and political and technical difficulties inhibit wireless technologies
- Wireless channel
  - Line-of-sight is best but not required
  - Signals can still be received
    - Transmission through objects
    - Reflections off of objects
    - Scattering of signals
    - Diffraction around edges of objects

The trouble with wireless

- Wireless channel
  - Reflections can cause multiple copies of the signal to arrive
    - At different times and attenuations
    - Creates the problem of *multipath fading*
    - Signals add together to degrade the final signal
  - Noise
  - Interference from other users
  - Doppler spread caused by movement

Combating Problems of Wirelesss

Combating problems:: Modulation

- **Modulation**
  - Map the digital data onto analog signals
  - Modulation is the process of changing a carrier signal to carry information (data).
  - Goal: Send more bits per signal.
  - In Wireless, the Carrier is a Sine Wave
  - To send data, we modulate:
    - Amplitude (A) → Amplitude Shift Keying (ASK)
    - Frequency (f) → Frequency Shift Keying (FSK)
    - Phase (ϕ) → Phase Shift Keying (PSK)
    - Or combinations → QAM (Quadrature Amplitude Modulation)
  - How This Works Visually: QPSK
    - Let's say we want to send 4 different symbols using QPSK:
    - We assign 00, 01, 10, 11 to four different phase angles of the carrier signal.
    - These are all sine waves of the same frequency, but with different starting points (phase shifts).

Combating problems:: Modulation

- **Modulation**
  Visual Analogy:
  - Imagine each subcarrier is a musical note 🎵.
  - You can make it louder/quieter (amplitude) or change the timing (phase) to send secret messages — that's modulation!

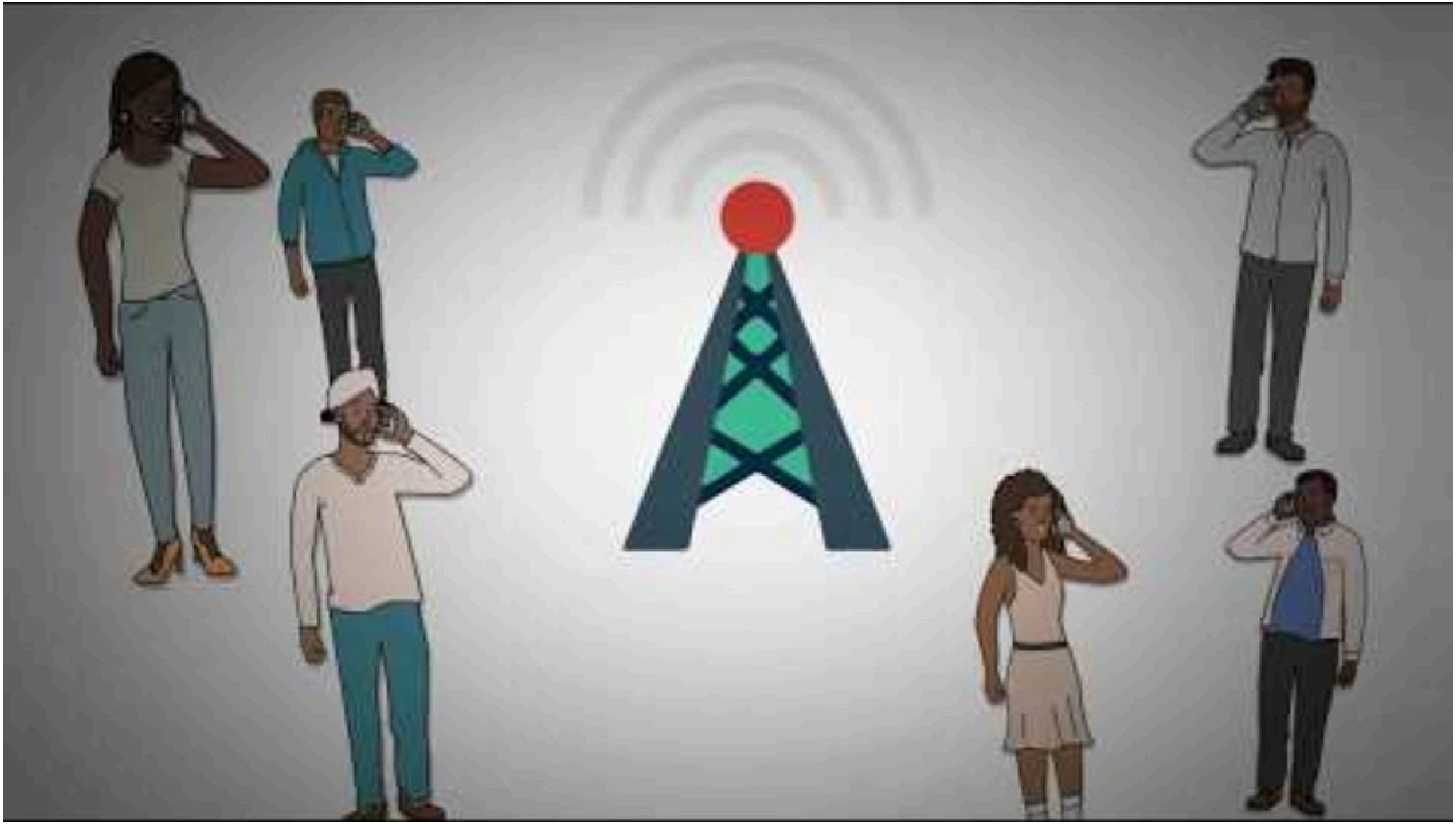
Concept	In Modulation
Input	Digital bits (e.g., 1010...)
Carrier wave	Sine wave at a certain frequency
Output	Modified wave carrying data
Visualization	Shift in wave’s amplitude/phase/frequency

Combating problems:: Error Coding & Equalization

- **Error Control Coding**
  - Adds **extra bits** to detect and fix errors.
  - *Example*: Like adding a checksum or parity bits in your data so you can spot and fix mistakes.
- **Adaptive Modulation and Coding**
  - **Dynamically adjusts** modulation/coding based on signal quality.
  - *Example*: If the signal is weak, switch to safer but slower mode; if strong, go faster.
- **Equalization**
  - Reduces effects of **multipath**, where signals bounce and cause distortion.
  - *Example*: Like tuning your ear to separate overlapping echoes in a large hall.

Combating problems:: Multiplexing

- **Multiplexing & Multiple Access**



Multiplexing

- A communication medium (like a cable or channel) can usually carry **more data** than just **one signal**.
- So, we use **multiplexing** to send **multiple signals at the same time** over that medium.
- This makes **better use** of the channel and **saves resources**.

2.11 Multiplexing

Multiplexing

- Frequency-division multiplexing (FDM)
  - Takes advantage of the fact that the useful bandwidth of the medium exceeds the required bandwidth of a given signal
- Time-division multiplexing (TDM)
  - Takes advantage of the fact that the achievable bit rate of the medium exceeds the required data rate of a digital signal

Combating problems:: Multiplexing

- **Multiplexing**
- What it does: Combines multiple data streams for transmission on the same physical medium.
- Types:
  - Time Division Multiplexing (TDM)
  - Frequency Division Multiplexing (FDM)
  - Code Division Multiplexing (CDM)
  - Space Division Multiplexing (SDM)
- Where: implemented at the physical layer (but different function than modulation).
- Analogy: Like scheduling multiple speakers to use a single microphone one after another (TDM) or letting them speak at different pitches (FDM).

- **Multiple Access**
- Multiple Access refers to how multiple users share the same communication medium simultaneously — it's about who gets to transmit when and how.

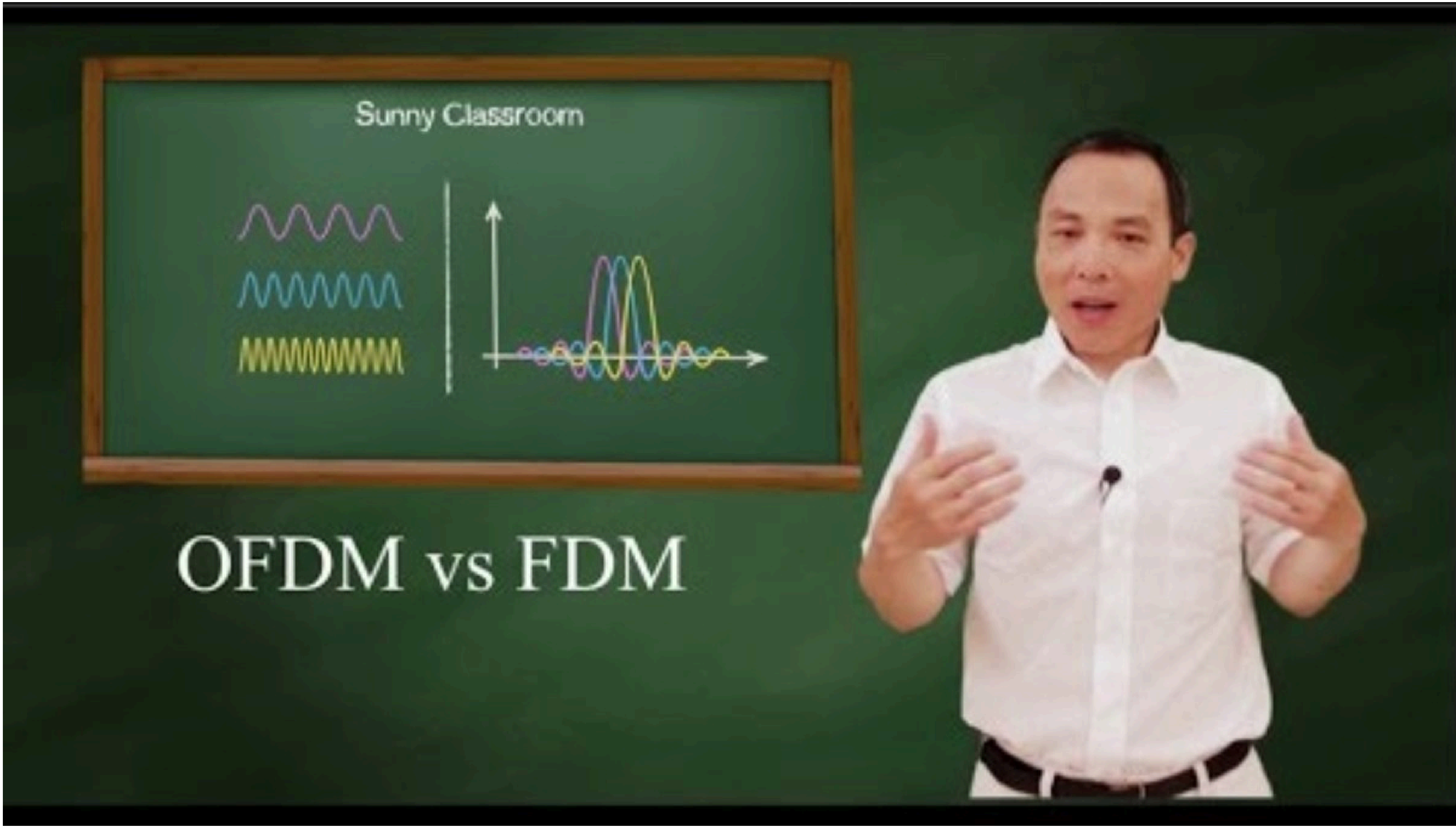
2.12 FDM and TDM

Reasons for Widespread Use of Multiplexing

- Cost per kbps of transmission facility declines with an increase in the data rate
- Cost of transmission and receiving equipment declines with increased data rate
- Most individual data communicating devices require relatively modest data rate support

OFDM
(Orthogonal Frequency Division Multiplexing)

Combating problems:: **OFDM**



Combating problems:: **OFDM**

Combating problems:: **OFDM**

Combating problems:: **OFDM**

OFDM (Orthogonal Frequency Division Multiplexing):

- A signal is divided into multiple low-rate data streams
- Transmitted simultaneously over closely spaced orthogonal subcarriers, reducing interference and multipath effects.

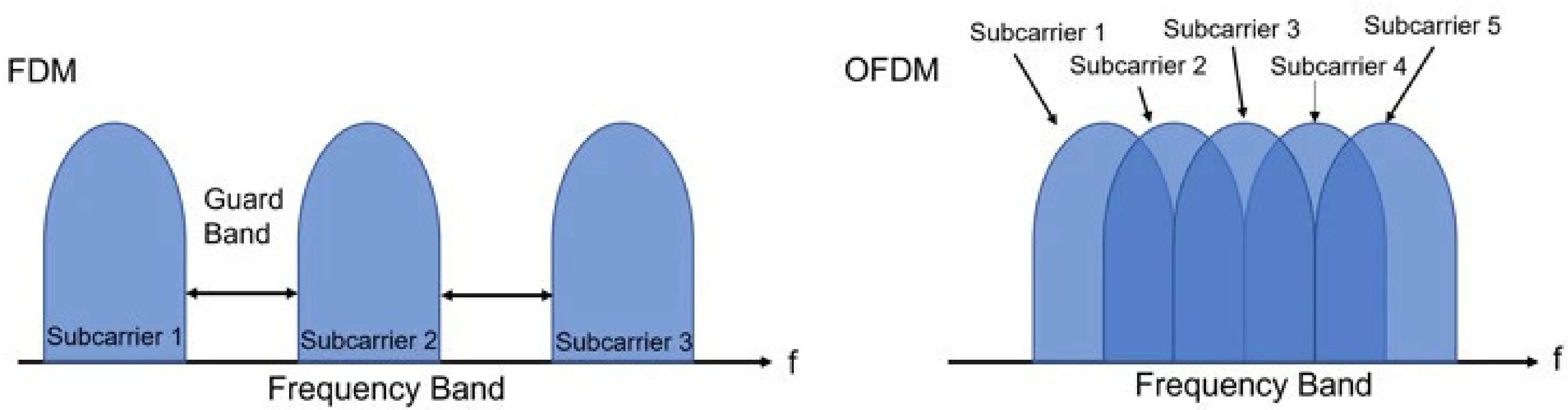


Figure: Conceptual diagram of multi-carrier transmission

Combating problems:: **OFDM**

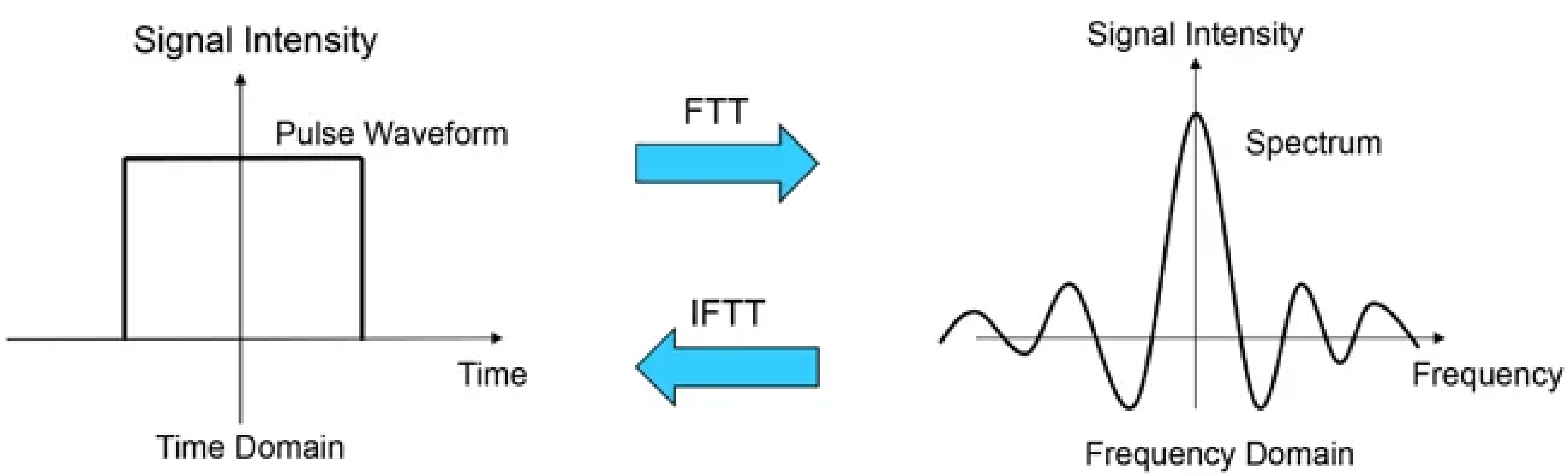
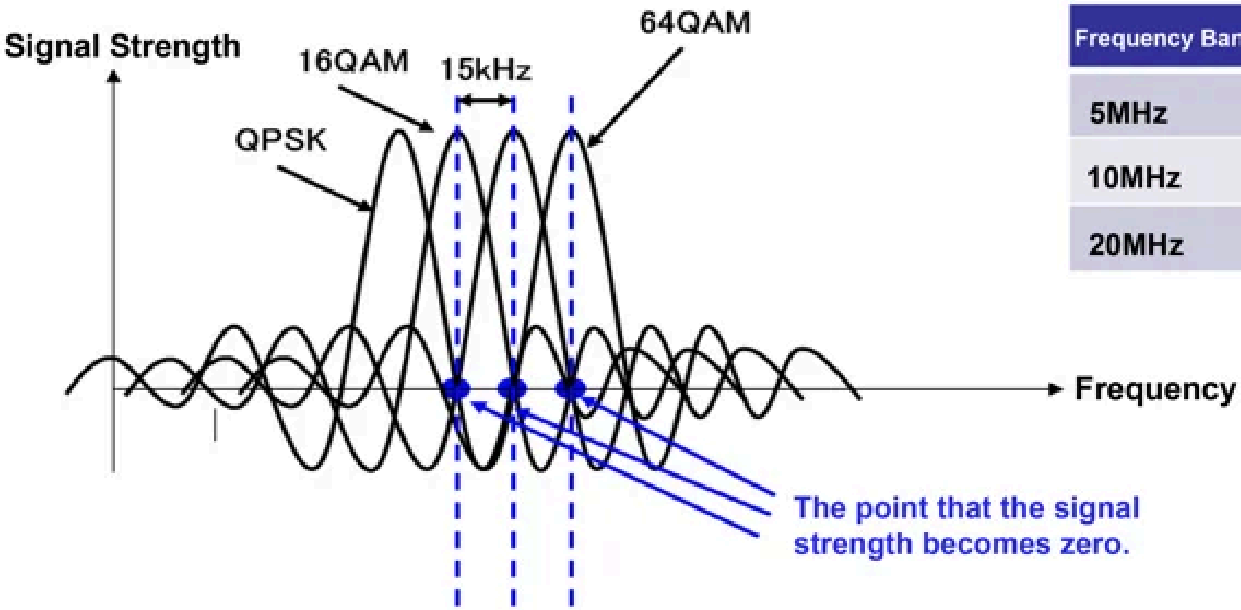


Figure: Representation of signals on the time and frequency axes

Combating problems:: **OFDM**

OFDM (Orthogonal Frequency Division Multiplexing):

- A signal is divided into multiple low-rate data streams
- Transmitted simultaneously over closely spaced orthogonal subcarriers, reducing interference and multipath effects.



Frequency Bandwidth	Subcarrier Number
5MHz	300
10MHz	600
20MHz	1200

Figure: Subcarrier arrangement in OFDM

- [Explore : For More Detail's of OFDM](#)

Combating problems:: **OFDM**

OFDM – Dividing Signal into Low-Rate Streams:

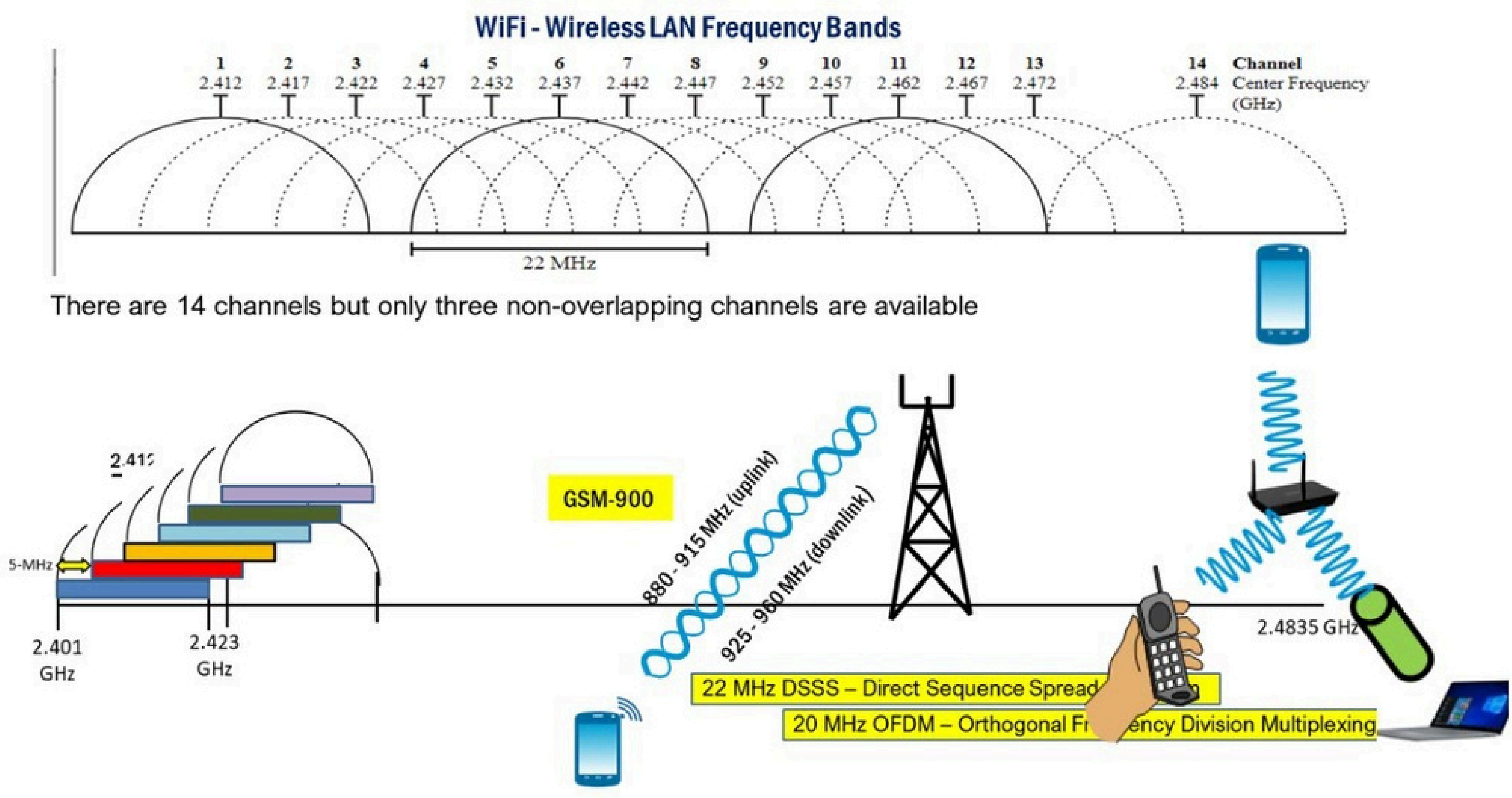
- Suppose you want to send a **fast stream of 1000 bits/sec**.
- Instead of sending it as **one fast stream**, OFDM splits it into **10 slower streams**, each sending **100 bits/sec**.
- These streams are then sent **simultaneously** on different sub-frequencies (subcarriers), which don’t interfere with each other because they are **orthogonal**.

  Simple Analogy:
- Imagine 10 people (subcarriers) speaking **slowly** at the same time, each in a **different room** (frequency).
- Even though each speaks slowly, **together** they deliver the full message **quickly and without interference**.
- So:

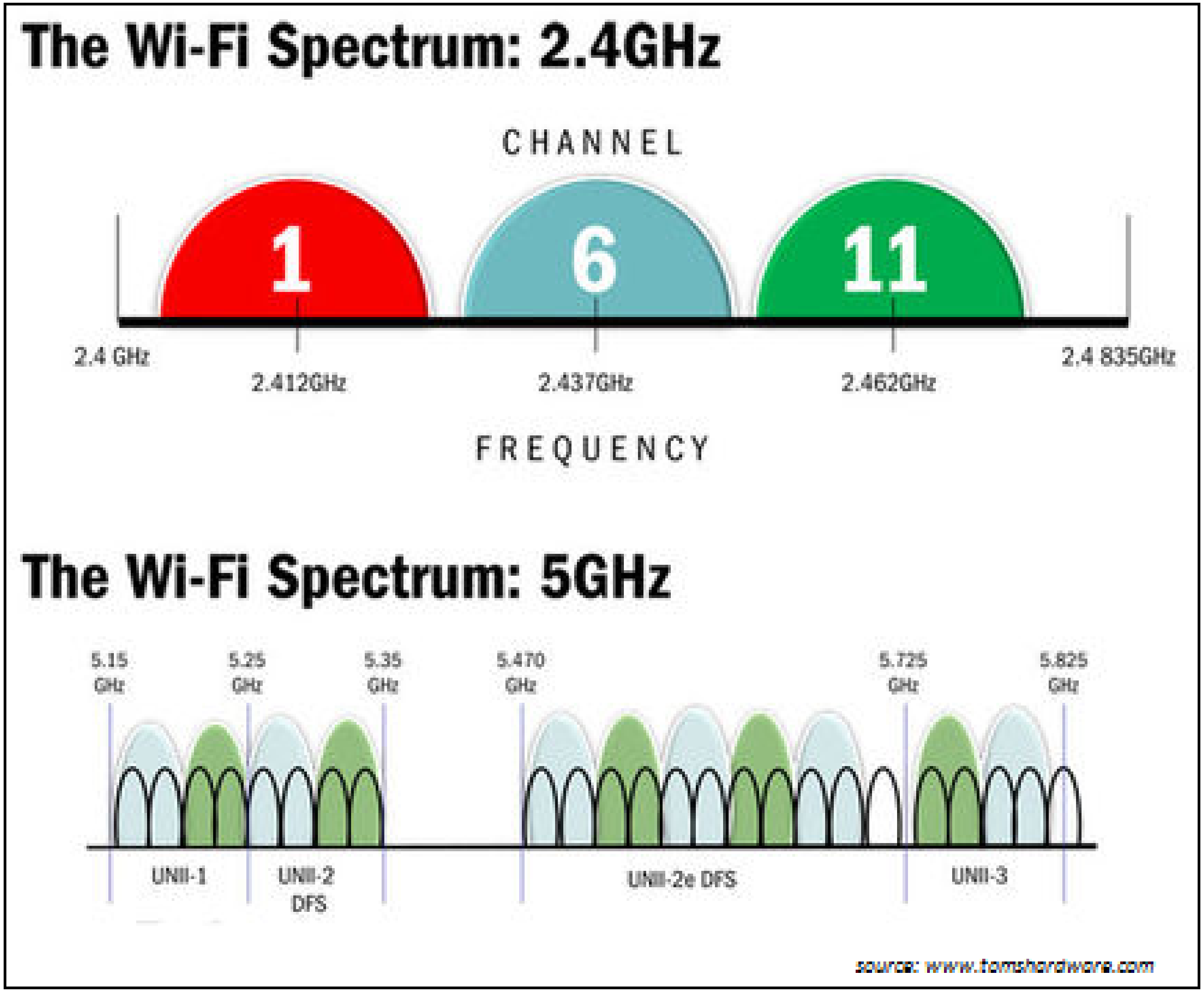
  “Low-rate data streams” = smaller, slower chunks of the original data

  Sent in parallel over **multiple frequencies** = faster, reliable transmission without overlap.

Radio Spectrum:: WiFi Channels



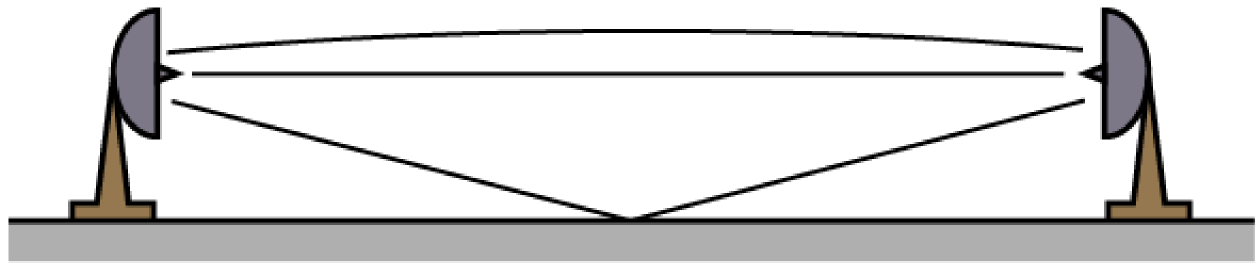
Radio Spectrum:: WiFi Channels



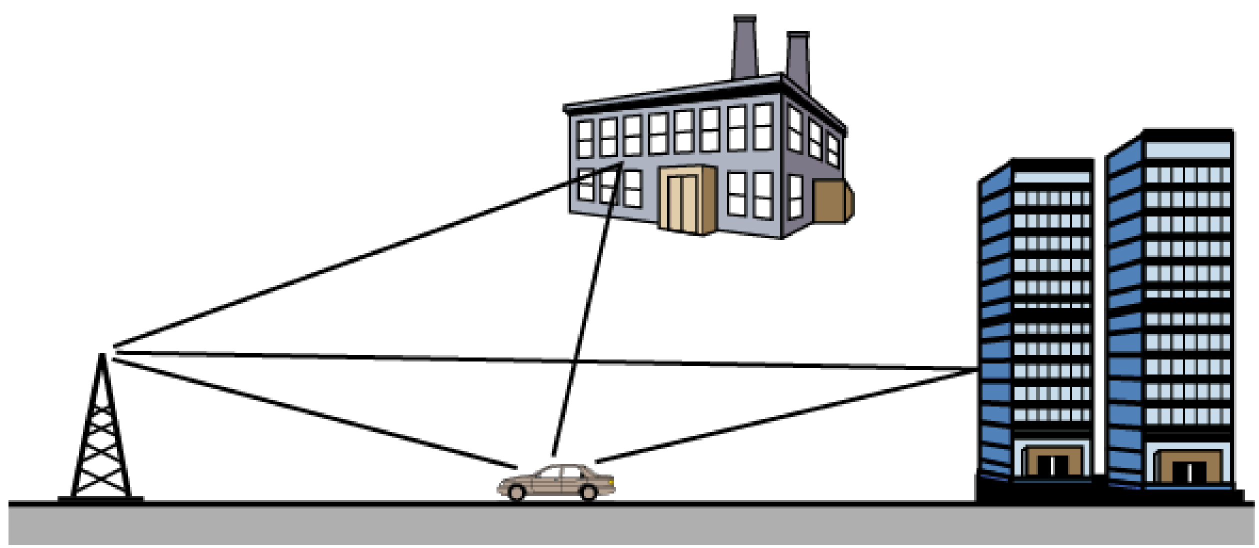
Combating problems..

MIMO (Multiple-Input Multiple-Output)

Multipath Propagation



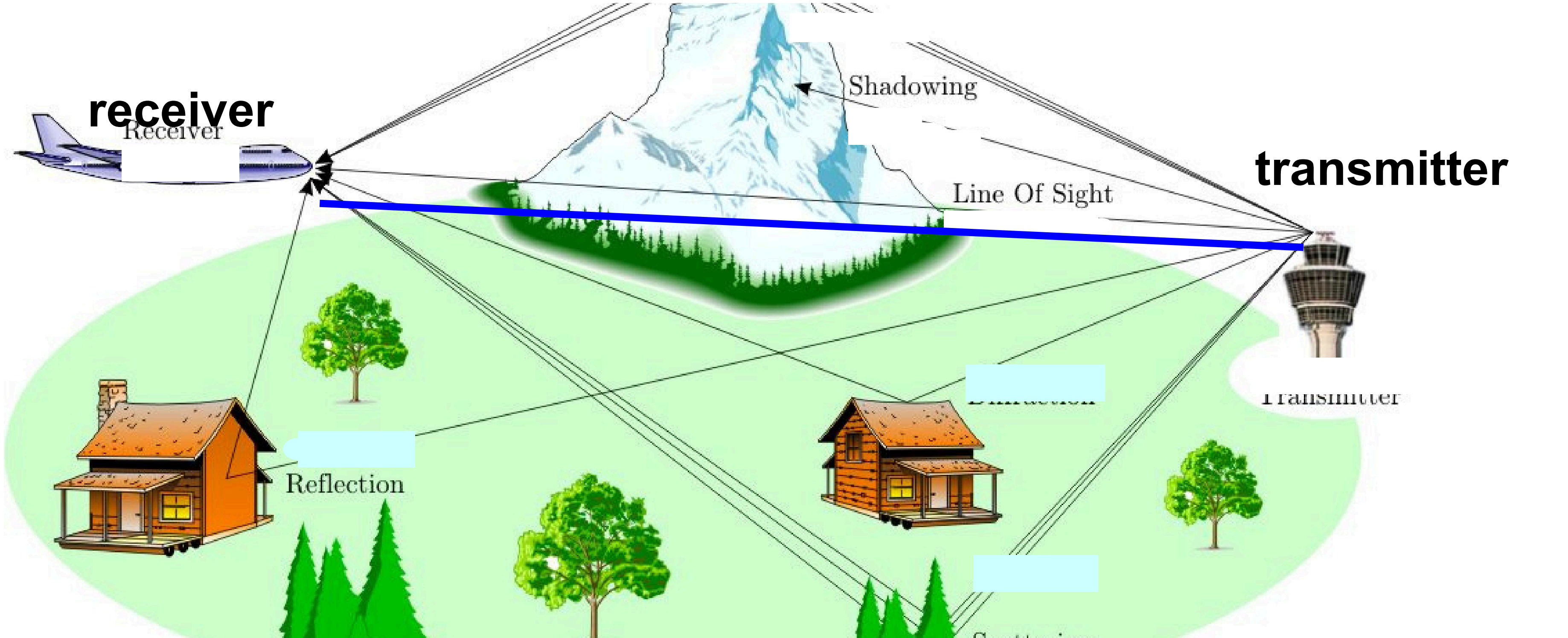
(a) Microwave line of sight



(b) Mobile radio

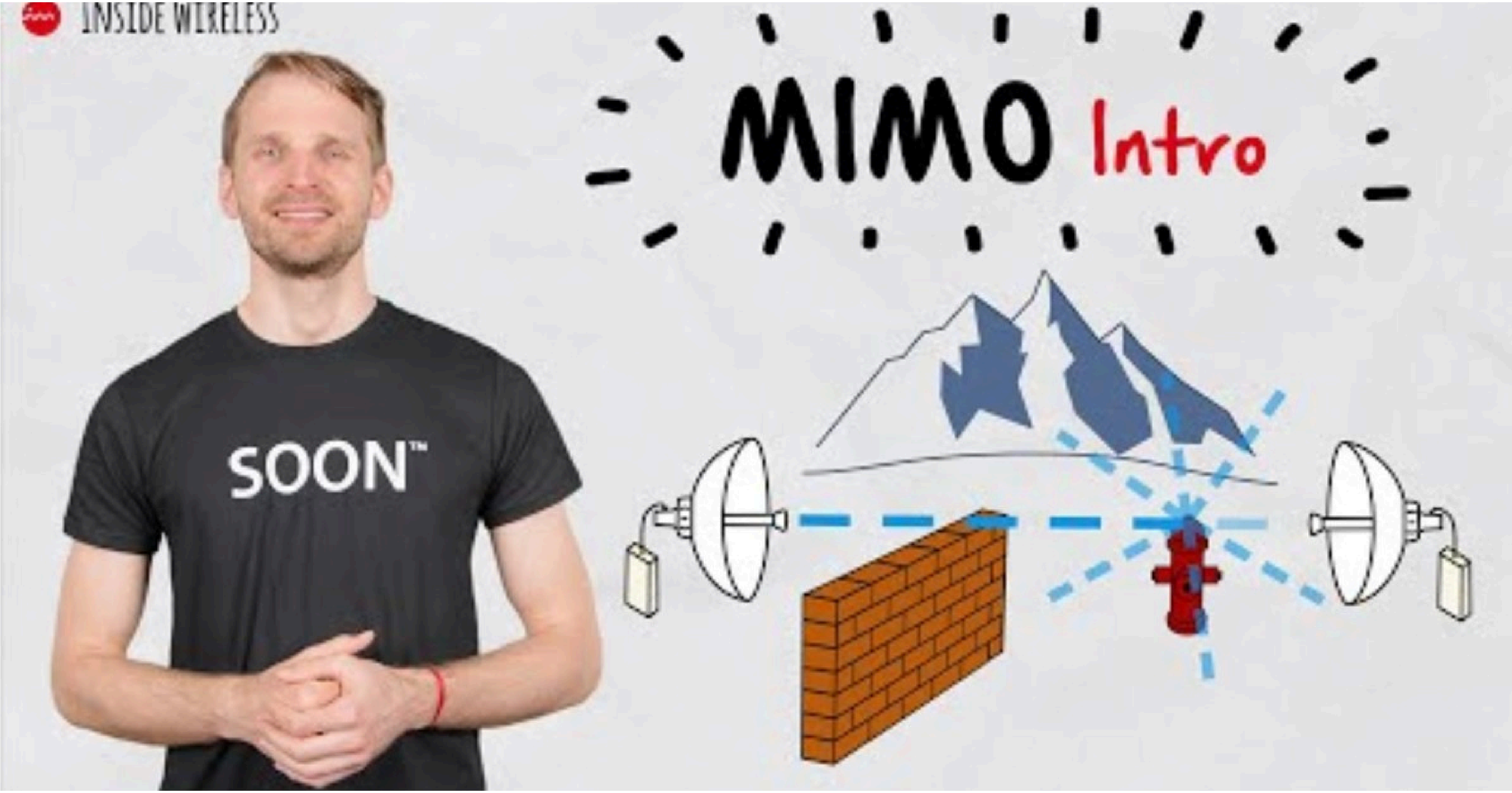
Multipath Propegation & Environmental Obstructions

- Multi-path propagation
  - Electromagnetic waves reflect off objects
  - Taking many paths of different lengths
  - Causing blurring of signal at the receiver



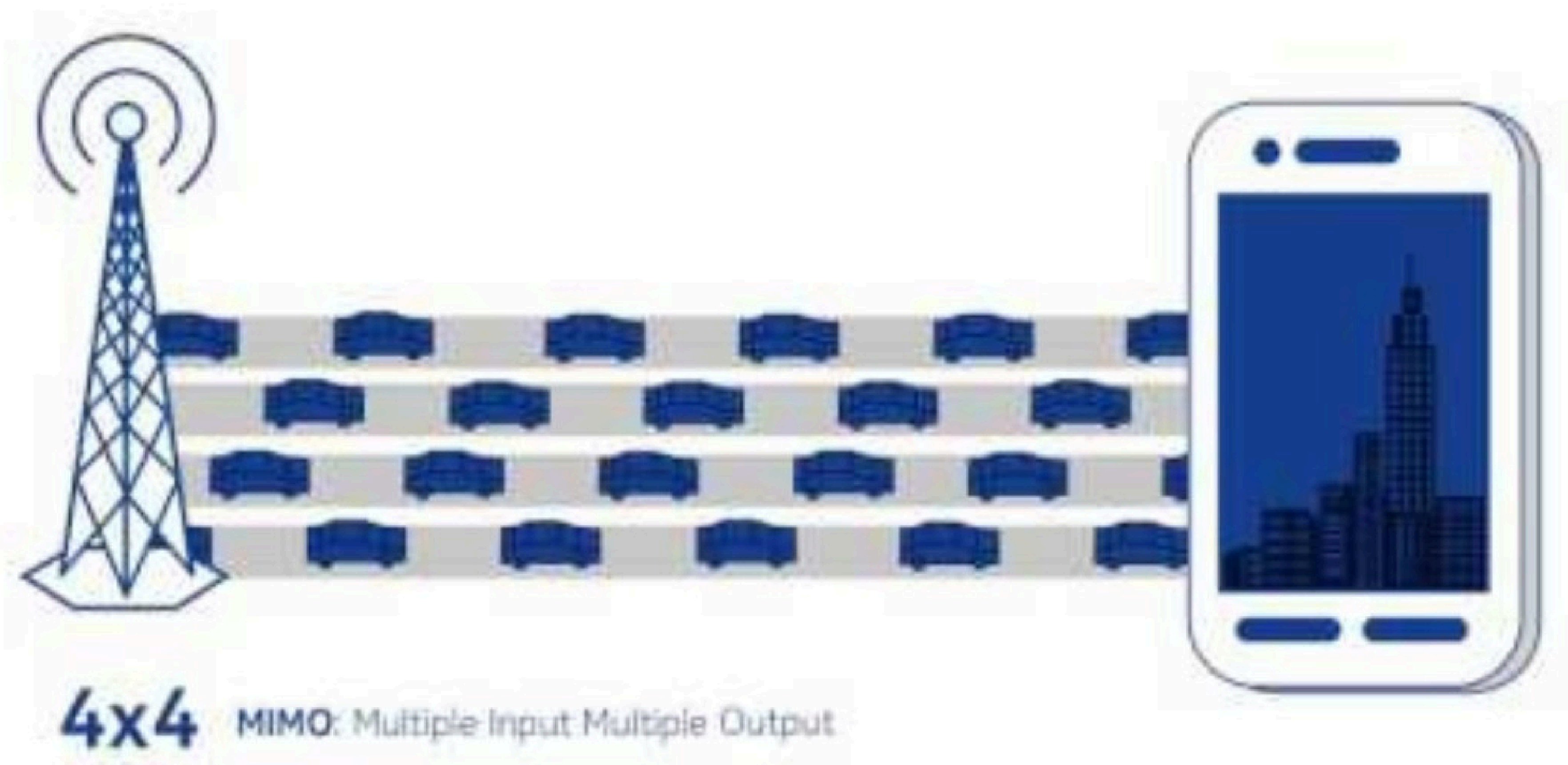
Combating problems:: MIMO

- **MIMO (Multiple-Input Multiple-Output)**



Combating problems:: MIMO

- **MIMO (Multiple-Input Multiple-Output)**



Combating problems:: MIMO

- **MIMO (Multiple-Input Multiple-Output)**
  - Uses multiple antennas to:
    - **Send multiple data streams at once** (parallel data)
    - 10s or 100s of antennas to focus antenna beams toward intended devices
  - *Example*: Like using several delivery trucks instead of one to send more packages.
- Multiple antennas **at the transmitter** send **multiple data streams simultaneously** over the **same frequency channel** !
  - **But how is that possible?**
  - It works by exploiting **spatial diversity** and **multipath propagation** in the environment.
  - There are two core techniques involved:
    - **1. Beamforming**
    - **2. Spatial Multiplexing**

Combating problems:: MIMO

- **MIMO (Multiple-Input Multiple-Output)**

1. **Beamforming**

- Uses **phase shifts** across antennas to **steer** the signal energy in **specific angular directions**.
- This is useful for **boosting signal strength** in a specific direction, or **reducing interference**.
- Common in **massive MIMO** (e.g., 5G base stations).

Combating problems:: MIMO

- **MIMO (Multiple-Input Multiple-Output)**

2. Spatial Multiplexing

- Different antennas transmit **different data streams** at the **same time** and on the **same frequency**.
- The receiver, using multiple antennas and smart signal processing (like **channel estimation**), can **separate** these overlapping signals.

- This **increases data rate** without needing more bandwidth.
- Direction isn't explicitly controlled — instead, it relies on **natural differences in path** (due to reflections, angles, etc.).

Combating problems:: MIMO

Spatial Multiplexing:

- Each transmitting antenna sends a **different data stream** at the **same frequency**.
- Due to the different locations of the transmitting and receiving antennas, and the **multipath environment**, the signal from each antenna takes **slightly different paths** and gets **mixed differently** at each receiver antenna.
- The receiver uses this **unique "mixture" of signals** to **separate the original streams** using signal processing algorithms.

Analogy:

- Imagine 3 people (antennas) shouting different words in a canyon.
- The echoes bounce differently to each of 3 listeners (receiver antennas) due to the shape of the rocks (environment).
- Even though they all shouted at once, clever listeners can figure out who said what by listening from different spots and **analyzing the echo patterns**.

Combating problems::

OFDM vs MIMO

Similarity:

- Both aim to **increase data rate and reliability**.
- Both are used **together** in modern systems like Wi-Fi 6 and 5G.
- Both support **parallelism** to handle high-speed data efficiently.

Dissimilarity:

Aspect	**OFDM**	**MIMO**
Transmission medium	Same antenna, many frequencies	Many antennas, same frequency
Focus	Handles frequency interference (multipath)	Handles spatial diversity/multipath
Hardware need	Can work with 1 antenna	Needs multiple antennas

Simple Analogy:

- **OFDM** is like sending a message split across **many radio stations** (frequencies) at once
- **MIMO** is like sending **different messages from different people (antennas)** all at the same time

Combating problems::

Spread Spectrum Techniques

Combating problems:: Spread Spectrum Techniques

Common Purpose:
To transmit data more reliably and securely by using more bandwidth than the minimum required.

1. **Frequency Hopping Spread Spectrum (FHSS)**
2. **Direct Sequence Spread Spectrum (DSSS)**

These are commonly used in **wireless communication systems**, especially where interference and security are concerns.
s
Why this name?
"spread" the original data signal over a **wider frequency band** than necessary for transmission. This helps in:

- **Reducing interference**
- **Improving security**
- **Resisting jamming**
- **Enhancing signal robustness**

Combating problems:: Spread Spectrum Techniques

Frequency Hopping Spread Spectrum (FHSS)

Idea:

The signal **jumps between different frequencies** very fast in a pattern known to both sender and receiver.

Simple Example:

Imagine two people talking using **walkie-talkies**, but they **change channels every second** following a shared schedule. Anyone trying to listen without knowing the pattern hears only random noise.

- Helps avoid interference and jamming.
- Used in Bluetooth, military radios.

Combating problems:: Spread Spectrum Techniques

Direct Sequence Spread Spectrum (DSSS)

Idea:

Each bit of the signal is replaced by a **longer code (spreading code)** that spreads it across a wide frequency range.

Simple Example:

Suppose we want to send 1011. DSSS would spread each bit using a long PN sequence like 110101001.

So, the transmitted signal becomes much longer but more robust.

Even if some bits are lost due to noise, the receiver can still figure out the original data.

- More resistant to noise and eavesdropping.
- Used in older Wi-Fi (802.11b), GPS.

References

- Chapter 2
- Chapter 5
- Chapter 6
- Chapter 7
- Chapter 8
- Chapter 9
    - Wireless Communication Networks and Systems by Beard and Stallings